

STEAM-AI Diagnostic Report

BASE_2025, horizon 2025, run BASE_2025_2025_52d15c27

Generated 2026-09-22T22:25:46+00:00 by STEAM-AI 0.1.0. Ingested 2026-09-22T21:06:03.422845+00:00.

DRAFT: DRAFT - generated by STEAM-AI for modeller review

SYNTHETIC DATA: SYNTHETIC DATA - this run was generated for testing, not a STEAM output

Run details

Item	Value
Run id	BASE_2025_2025_52d15c27
Scenario	BASE_2025
Horizon year	2025
Policy set	REF
STEAM version	synthetic-0.1
Ingested at	2026-09-22T21:06:03.422845+00:00
Report generated at	2026-09-22T22:25:46+00:00
STEAM-AI version	0.1.0

[src: manifest.json]

This report lists what STEAM-AI found when it checked the run, ranked by how much each issue matters. Every number carries a source reference in small grey text. The report is a draft for modeller review and is not a model result.

Executive summary

Run health score: 24.0 out of 100, grade E.

[src: health.json]

Definition: Health = 0.7 x Findings component + 0.3 x Convergence component. Findings component = $\max(0, 100 - 25 \times \text{Critical} - 8 \times \text{High} - 2 \times \text{Medium} - 0 \times \text{Info})$, counting each distinct problem type (a check and the issue it found) once, at its worst significant severity: a coding error repeated on 8,000 links is one problem. Convergence component = 100 if every assignment period met the relative gap target, scaled down linearly to 0 at ten times the target. When a run has no convergence data, the findings component carries the full weight. Grade: A ≥ 90 , B ≥ 75 , C ≥ 60 , D ≥ 40 , E below 40.

[src: config/health.yaml; health.json]

Health score components

Component	Score	Weight	Detail
Findings	0.0	0.70	100 - 156 penalty over 14 distinct problem type(s) across 40 significant finding(s) (Critical 14, High 13, Medium 13)
Convergence	80.0	0.30	target 0.001; per period: HWY_ASSIGN AM: gap 0.0200 -> 0; HWY_ASSIGN EV: gap 0.0005 -> 100; HWY_ASSIGN MD: gap 0.0005 -> 100; HWY_ASSIGN NT: gap 0.0005 -> 100; HWY_ASSIGN PM: gap 0.0005 -> 100

[src: health.json]

The checks produced 40 findings: 14 Critical, 13 High, 13 Medium, 0 Info.

[src: findings.json]

Findings by severity

Severity	Findings	Significant
Critical	14	14
High	13	13
Medium	13	13
Info	0	0

[src: findings.json]

Top five findings

Top five findings by severity

Rank	Severity	Location	Finding
1	Critical	Link 5 (ART, 1003-1004)	Link 5 (ART) has a capacity of 0 vehicles per hour, so traffic cannot realistically use it.
2	Critical	Link 199 (FWY, 1107-1108), AM	Link 199 (FWY) carries 28,080 vehicles in the AM peak, 1.56 times its capacity.
3	Critical	Link 191 (FWY, 1103-1104), AM	Link 191 (FWY) carries 26,640 vehicles in the AM peak, 1.48 times its capacity.
4	Critical	Link 193 (FWY, 1104-1105), AM	Link 193 (FWY) carries 27,000 vehicles in the AM peak, 1.50 times its capacity.
5	Critical	Link 189 (FWY, 1102-1103), AM	Link 189 (FWY) carries 26,280 vehicles in the AM peak, 1.46 times its capacity.

[src: findings/1a0b9ff0e61f; findings/24992b6a31d7; findings/3cfc28052b56; findings/3e5c6f481bed; findings/65aa8257d2d8]

What to fix first

27 Critical or High findings need action. Fix them in the order below; Critical items block use of the run.

[src: findings.json]

Action list

Severity	Location	Finding	Suggested action	Best measure
Critical	Link 5 (ART, 1003-1004)	Link 5 (ART) has a capacity of 0 vehicles per hour, so traffic cannot realistically use it.	Fix the link attributes or remove the link from the scenario network.	Repair the network coding (not computable, low confidence)
Critical	Link 199 (FWY, 1107-1108), AM	Link 199 (FWY) carries 28,080 vehicles in the AM peak, 1.56 times its capacity.	Check lanes and capacity coding on this link and its competitors; compare with counts if available.	Add one lane (sketch elasticity, medium confidence): V/C falls from 1.56 to about 1.17 if demand is unchanged
Critical	Link 191 (FWY, 1103-1104), AM	Link 191 (FWY) carries 26,640 vehicles in the AM peak, 1.48 times its capacity.	Check lanes and capacity coding on this link and its competitors; compare with counts if available.	Add one lane (sketch elasticity, medium confidence): V/C falls from 1.48 to about 1.11 if demand is unchanged
Critical	Link 193 (FWY, 1104-1105), AM	Link 193 (FWY) carries 27,000 vehicles in the AM peak, 1.50 times its capacity.	Check lanes and capacity coding on this link and its competitors; compare with counts if available.	Add one lane (sketch elasticity, medium confidence): V/C falls from 1.50 to about 1.12 if demand is unchanged
Critical	Link 189 (FWY, 1102-1103), AM	Link 189 (FWY) carries 26,280 vehicles in the AM peak, 1.46 times its capacity.	Check lanes and capacity coding on this link and its competitors; compare with counts if available.	Add one lane (sketch elasticity, medium confidence): V/C falls from 1.46 to about 1.09 if demand is unchanged
Critical	Link 197 (FWY, 1106-1107), AM	Link 197 (FWY) carries 27,720 vehicles in the AM peak, 1.54 times its capacity.	Check lanes and capacity coding on this link and its competitors; compare with counts if available.	Add one lane (sketch elasticity, medium confidence): V/C falls from 1.54 to about 1.16 if demand is unchanged
Critical	POP / ALL	Total population across all zones is 283,630, 10.7% below the control total of 317,666.	Scale POP across all zones by 1.120 to match the control total, or confirm the control total is current.	Scale POP by 1.120 (sketch elasticity, high confidence): total becomes 317,666; trip ends move roughly in proportion
Critical	Link 185 (FWY, 1100-1101), AM	Link 185 (FWY) carries 25,560 vehicles in the AM peak, 1.42 times its capacity.	Check lanes and capacity coding on this link and its competitors; compare with counts if available.	Add one lane (sketch elasticity, medium confidence): V/C falls from 1.42 to about 1.06 if demand is unchanged
Critical	Link 201 (FWY, 1108-1109), AM	Link 201 (FWY) carries 28,440 vehicles in the AM peak, 1.58 times its capacity.	Check lanes and capacity coding on this link and its competitors; compare with counts if available.	Add one lane (sketch elasticity, medium confidence): V/C falls from 1.58 to about 1.19 if demand is unchanged
Critical	Link 6 (ART, 1004-1003)	Link 6 (ART) has a capacity of 0 vehicles per hour, so traffic cannot realistically use it.	Fix the link attributes or remove the link from the scenario network.	Repair the network coding (not computable, low confidence)

[src: findings/1a0b9ff0e61f; findings/24992b6a31d7; findings/3cfc28052b56; findings/3e5c6f481bed; findings/65aa8257d2d8; findings/81d9e99d5ab0; findings/8e85b004d345; findings/95aff43d2c28; findings/9914fb4f0d20; findings/ad89fdca2740]

17 further Critical or High findings are listed in the findings section.

[src: findings.json]

Findings by severity

Critical findings (14)

Network connectivity: Link 5 (ART, 1003-1004)

Critical: Link 5 (ART) has a capacity of 0 vehicles per hour, so traffic cannot realistically use it.

Modeller view: Method: attribute threshold on links

Values: issue=zero_capacity, capacity_vph=0, ffs_kph=60, lanes=2, link_class=ART

Thresholds: min_capacity_vph=1, min_ffs_kph=5

Files: links.csv rows 5

SQL: SELECT link_id, link_class, capacity_vph, ffs_kph, lanes, source_file, source_row FROM links WHERE capacity_vph < 1.0 OR ffs_kph < 5.0 ORDER BY link_id

[src: findings/1a0b9ff0e61f; links.csv (links) row 5 col capacity_vph]

Likely cause: Capacity or speed lookup missed this link class/area type, or the link was meant to be closed.

[src: findings/1a0b9ff0e61f; links.csv (links) row 5 col capacity_vph]

Suggested action: Fix the link attributes or remove the link from the scenario network.

[src: findings/1a0b9ff0e61f; links.csv (links) row 5 col capacity_vph]

Evidence

Metric	Value	Threshold
issue	zero_capacity	
capacity_vph	0	
ffs_kph	60	
lanes	2	
link_class	ART	
min_capacity_vph		0
min_ffs_kph		5

[src: findings/1a0b9ff0e61f; links.csv (links) row 5 col capacity_vph]

Sources: links.csv (links) row 5 col capacity_vph

[src: findings/1a0b9ff0e61f; links.csv (links) row 5 col capacity_vph]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Repair the network coding	Fix the attribute or add the missing link/one-way direction and re-assign.	Not estimated	not computable	low

[src: findings/1a0b9ff0e61f; links.csv (links) row 5 col capacity_vph]

Link volume outliers: Link 199 (FWY, 1107-1108), AM

Critical: Link 199 (FWY) carries 28,080 vehicles in the AM peak, 1.56 times its capacity.

Modeller view: Method: V/C from link_flows; robust z = (ln(v+1) - median) / (1.4826 MAD) within link_class x area_type x period

Values: issue=over_capacity, volume=2.808e+04, vc_ratio=1.56, zscore=8.476,

group_median_volume=6796, group_size=24, link_class=FWY, area_type=CBD, capacity_vph=6000, lanes=3, too_low=False

Thresholds: vc_critical=1.3, vc_high=1.1, vc_medium=0.95, zscore_threshold=3, min_group_size=20, low_volume_share_of_group_median=0.02, min_vc_for_outlier=0.25

Files: link_flows.csv rows 3103

SQL: WITH ph(period, hours) AS (VALUES ('AM', 3.0), ('PM', 3.0)), f AS (SELECT f.link_id, f.period, f.volume, COALESCE(f.vc_ratio, f.volume / NULLIF(l.capacity_vph * ph.hours, 0)) AS vc_ratio, l.link_class, l.area_type, l.capacity_vph, l.lanes, f.source_file, f.source_row FROM link_flows f JOIN links l USING (link_id) JOIN ph ON ph.period = f.period WHERE f.user_class = 'ALL' AND f.volume IS NOT NULL AND l.link_class NOT IN ('CONN')), g AS (SELECT link_class, area_type, period, COUNT(*) AS n, MEDIAN(LN(volume + 1)) AS med_log, MEDIAN(volume) AS med_vol FROM f GROUP BY 1, 2, 3), d AS (SELECT f.*, g.n, g.med_log, g.med_vol, LN(f.volume + 1) - g.med_log AS dev FROM f JOIN g USING (link_class, area_type, period)), m AS (SELECT link_class, area_type, period, MEDIAN(ABS(dev)) AS mad FROM d GROUP BY 1, 2, 3), scored AS (SELECT d.*, m.mad, CASE WHEN d.n >= 20 AND m.mad > 0 THEN d.dev / (1.4826 * m.mad) END AS zscore, (d.n >= 20 AND d.capacity_vph > 0 AND d.med_vol > 0 AND d.volume < 0.02 * d.med_vol) AS too_low FROM d JOIN m USING (link_class, area_type, period)) SELECT * FROM scored WHERE vc_ratio >= 0.95 OR (ABS(zscore) >= 3.0 AND COALESCE(vc_ratio, 0) >= 0.25) OR too_low ORDER BY vc_ratio DESC NULLS LAST, ABS(zscore) DESC NULLS LAST

[src: findings/24992b6a31d7; link_flows.csv (link_flows) row 3103 col volume]

Likely cause: Demand concentrated on this link because parallel routes are missing or slow, or the capacity/lanes are under-coded.

[src: findings/24992b6a31d7; link_flows.csv (link_flows) row 3103 col volume]

Suggested action: Check lanes and capacity coding on this link and its competitors; compare with counts if available.

[src: findings/24992b6a31d7; link_flows.csv (link_flows) row 3103 col volume]

Evidence

Metric	Value	Threshold
issue	over_capacity	
volume	28,080	
vc_ratio	1.56	
zscore	8.476	
group_median_volume	6,800	
group_size	24	
link_class	FWY	
area_type	CBD	
capacity_vph	6,000	
lanes	3	
too_low	no	
vc_critical		1.30
vc_high		1.10
vc_medium		0.95
zscore_threshold		3
min_group_size		20
low_volume_share_of_group_median		2.0%
min_vc_for_outlier		0.25

[src: findings/24992b6a31d7; link_flows.csv (link_flows) row 3103 col volume]

Sources: link_flows.csv (link_flows) row 3103 col volume

[src: findings/24992b6a31d7; link_flows.csv (link_flows) row 3103 col volume]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Add one lane	Add a lane to this link, raising capacity from 6,000 to 8,000 veh/h (one lane = 2,000 veh/h).	V/C falls from 1.56 to about 1.17 if demand is unchanged	sketch elasticity	medium
Verify capacity coding	Check that lanes and capacity per lane match the link's real cross-section; an under-coded capacity produces the same symptom as real congestion.	Not estimated	not computable	medium
Check for missing parallel routes	Confirm that parallel roads in the corridor are coded and open in this scenario; a closed or missing alternative concentrates demand here.	Not estimated	not computable	low

[src: findings/24992b6a31d7; link_flows.csv (link_flows) row 3103 col volume]

Link volume outliers: Link 191 (FWY, 1103-1104), AM

Critical: Link 191 (FWY) carries 26,640 vehicles in the AM peak, 1.48 times its capacity.

Modeller view: Method: V/C from link_flows; robust $z = (\ln(v+1) - \text{median}) / (1.4826 \text{ MAD})$ within link_class x area_type x period

Values: issue=over_capacity, volume=2.664e+04, vc_ratio=1.48, zscore=8.161, group_median_volume=6796, group_size=24, link_class=FWY, area_type=CBD, capacity_vph=6000, lanes=3, too_low=False

Thresholds: vc_critical=1.3, vc_high=1.1, vc_medium=0.95, zscore_threshold=3, min_group_size=20, low_volume_share_of_group_median=0.02, min_vc_for_outlier=0.25

Files: link_flows.csv rows 3095

SQL: WITH ph(period, hours) AS (VALUES ('AM', 3.0), ('PM', 3.0)), f AS (SELECT f.link_id, f.period, f.volume, COALESCE(f.vc_ratio, f.volume / NULLIF(l.capacity_vph * ph.hours, 0)) AS vc_ratio, l.link_class, l.area_type, l.capacity_vph, l.lanes, f.source_file, f.source_row FROM link_flows f JOIN links l USING (link_id) JOIN ph ON ph.period = f.period WHERE f.user_class = 'ALL' AND f.volume IS NOT NULL AND l.link_class NOT IN ('CONN')), g AS (SELECT link_class, area_type, period, COUNT(*) AS n, MEDIAN(LN(volume + 1)) AS med_log, MEDIAN(volume) AS med_vol FROM f GROUP BY 1, 2, 3), d AS (SELECT f.*, g.n, g.med_log, g.med_vol, LN(f.volume + 1) - g.med_log AS dev FROM f JOIN g USING (link_class, area_type, period)), m AS (SELECT link_class, area_type, period, MEDIAN(ABS(dev)) AS mad FROM d GROUP BY 1, 2, 3), scored AS (SELECT d.*, m.mad, CASE WHEN d.n >= 20 AND m.mad > 0 THEN d.dev / (1.4826 * m.mad) END AS zscore, (d.n >= 20 AND d.capacity_vph > 0 AND d.med_vol > 0 AND d.volume < 0.02 * d.med_vol) AS too_low FROM d JOIN m USING (link_class, area_type, period)) SELECT * FROM scored WHERE vc_ratio >= 0.95 OR (ABS(zscore) >= 3.0 AND COALESCE(vc_ratio, 0) >= 0.25) OR too_low ORDER BY vc_ratio DESC NULLS LAST, ABS(zscore) DESC NULLS LAST

[src: findings/3cfc28052b56; link_flows.csv (link_flows) row 3095 col volume]

Likely cause: Demand concentrated on this link because parallel routes are missing or slow, or the capacity/lanes are under-coded.

[src: findings/3cfc28052b56; link_flows.csv (link_flows) row 3095 col volume]

Suggested action: Check lanes and capacity coding on this link and its competitors; compare with counts if available.

[src: findings/3cfc28052b56; link_flows.csv (link_flows) row 3095 col volume]

Evidence

Metric	Value	Threshold
issue	over_capacity	
volume	26,640	
vc_ratio	1.48	
zscore	8.161	
group_median_volume	6,800	

Metric	Value	Threshold
group_size	24	
link_class	FWY	
area_type	CBD	
capacity_vph	6,000	
lanes	3	
too_low	no	
vc_critical		1.30
vc_high		1.10
vc_medium		0.95
zscore_threshold		3
min_group_size		20
low_volume_share_of_group_median		2.0%
min_vc_for_outlier		0.25

[src: findings/3cfc28052b56; link_flows.csv (link_flows) row 3095 col volume]

Sources: link_flows.csv (link_flows) row 3095 col volume

[src: findings/3cfc28052b56; link_flows.csv (link_flows) row 3095 col volume]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Add one lane	Add a lane to this link, raising capacity from 6,000 to 8,000 veh/h (one lane = 2,000 veh/h).	V/C falls from 1.48 to about 1.11 if demand is unchanged	sketch elasticity	medium
Verify capacity coding	Check that lanes and capacity per lane match the link's real cross-section; an under-coded capacity produces the same symptom as real congestion.	Not estimated	not computable	medium
Check for missing parallel routes	Confirm that parallel roads in the corridor are coded and open in this scenario; a closed or missing alternative concentrates demand here.	Not estimated	not computable	low

[src: findings/3cfc28052b56; link_flows.csv (link_flows) row 3095 col volume]

Link volume outliers: Link 193 (FWY, 1104-1105), AM

Critical: Link 193 (FWY) carries 27,000 vehicles in the AM peak, 1.50 times its capacity.

Modeller view: Method: V/C from link_flows; robust $z = (\ln(v+1) - \text{median}) / (1.4826 \text{ MAD})$ within link_class x area_type x period

Values: issue=over_capacity, volume=2.7e+04, vc_ratio=1.5, zscore=8.241, group_median_volume=6796, group_size=24, link_class=FWY, area_type=CBD, capacity_vph=6000, lanes=3, too_low=False

Thresholds: vc_critical=1.3, vc_high=1.1, vc_medium=0.95, zscore_threshold=3, min_group_size=20, low_volume_share_of_group_median=0.02, min_vc_for_outlier=0.25

Files: link_flows.csv rows 3097

SQL: WITH ph(period, hours) AS (VALUES ('AM', 3.0), ('PM', 3.0)), f AS (SELECT f.link_id, f.period, f.volume, COALESCE(f.vc_ratio, f.volume / NULLIF(l.capacity_vph * ph.hours, 0)) AS vc_ratio, l.link_class, l.area_type, l.capacity_vph, l.lanes, f.source_file, f.source_row FROM link_flows f JOIN links l USING (link_id) JOIN ph ON ph.period = f.period WHERE f.user_class = 'ALL' AND f.volume IS NOT NULL AND l.link_class NOT IN ('CONN')), g AS (SELECT link_class, area_type, period, COUNT(*) AS n, MEDIAN(LN(volume + 1)) AS med_log, MEDIAN(volume) AS med_vol FROM f GROUP BY 1, 2, 3), d AS (SELECT f.*, g.n, g.med_log, g.med_vol, LN(f.volume + 1) - g.med_log AS dev FROM f JOIN g USING (link_class, area_type, period)), m AS (SELECT link_class, area_type, period, MEDIAN(ABS(dev)) AS

mad FROM d GROUP BY 1, 2, 3), scored AS (SELECT d.*, m.mad, CASE WHEN d.n >= 20 AND m.mad > 0 THEN d.dev / (1.4826 * m.mad) END AS zscore, (d.n >= 20 AND d.capacity_vph > 0 AND d.med_vol > 0 AND d.volume < 0.02 * d.med_vol) AS too_low FROM d JOIN m USING (link_class, area_type, period)) SELECT * FROM scored WHERE vc_ratio >= 0.95 OR (ABS(zscore) >= 3.0 AND COALESCE(vc_ratio, 0) >= 0.25) OR too_low ORDER BY vc_ratio DESC NULLS LAST, ABS(zscore) DESC NULLS LAST

[src: findings/3e5c6f481bed; link_flows.csv (link_flows) row 3097 col volume]

Likely cause: Demand concentrated on this link because parallel routes are missing or slow, or the capacity/lanes are under-coded.

[src: findings/3e5c6f481bed; link_flows.csv (link_flows) row 3097 col volume]

Suggested action: Check lanes and capacity coding on this link and its competitors; compare with counts if available.

[src: findings/3e5c6f481bed; link_flows.csv (link_flows) row 3097 col volume]

Evidence

Metric	Value	Threshold
issue	over_capacity	
volume	27,000	
vc_ratio	1.50	
zscore	8.241	
group_median_volume	6,800	
group_size	24	
link_class	FWY	
area_type	CBD	
capacity_vph	6,000	
lanes	3	
too_low	no	
vc_critical		1.30
vc_high		1.10
vc_medium		0.95
zscore_threshold		3
min_group_size		20
low_volume_share_of_group_median		2.0%
min_vc_for_outlier		0.25

[src: findings/3e5c6f481bed; link_flows.csv (link_flows) row 3097 col volume]

Sources: link_flows.csv (link_flows) row 3097 col volume

[src: findings/3e5c6f481bed; link_flows.csv (link_flows) row 3097 col volume]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Add one lane	Add a lane to this link, raising capacity from 6,000 to 8,000 veh/h (one lane = 2,000 veh/h).	V/C falls from 1.50 to about 1.12 if demand is unchanged	sketch elasticity	medium
Verify capacity coding	Check that lanes and capacity per lane match the link's real cross-section; an under-coded capacity produces the same symptom as real congestion.	Not estimated	not computable	medium
Check for missing parallel routes	Confirm that parallel roads in the corridor are coded and open in this scenario; a closed or missing alternative concentrates demand here.	Not estimated	not computable	low

[src: findings/3e5c6f481bed; link_flows.csv (link_flows) row 3097 col volume]

Link volume outliers: Link 189 (FWY, 1102-1103), AM

Critical: Link 189 (FWY) carries 26,280 vehicles in the AM peak, 1.46 times its capacity.

Modeller view: Method: V/C from link_flows; robust $z = (\ln(v+1) - \text{median}) / (1.4826 \text{ MAD})$ within link_class x area_type x period

Values: issue=over_capacity, volume=2.628e+04, vc_ratio=1.46, zscore=9.417, group_median_volume=5162, group_size=40, link_class=FWY, area_type=URB, capacity_vph=6000, lanes=3, too_low=False

Thresholds: vc_critical=1.3, vc_high=1.1, vc_medium=0.95, zscore_threshold=3, min_group_size=20, low_volume_share_of_group_median=0.02, min_vc_for_outlier=0.25

Files: link_flows.csv rows 3093

SQL: WITH ph(period, hours) AS (VALUES ('AM', 3.0), ('PM', 3.0)), f AS (SELECT f.link_id, f.period, f.volume, COALESCE(f.vc_ratio, f.volume / NULLIF(l.capacity_vph * ph.hours, 0)) AS vc_ratio, l.link_class, l.area_type, l.capacity_vph, l.lanes, f.source_file, f.source_row FROM link_flows f JOIN links l USING (link_id) JOIN ph ON ph.period = f.period WHERE f.user_class = 'ALL' AND f.volume IS NOT NULL AND l.link_class NOT IN ('CONN')), g AS (SELECT link_class, area_type, period, COUNT(*) AS n, MEDIAN(LN(volume + 1)) AS med_log, MEDIAN(volume) AS med_vol FROM f GROUP BY 1, 2, 3), d AS (SELECT f.*, g.n, g.med_log, g.med_vol, LN(f.volume + 1) - g.med_log AS dev FROM f JOIN g USING (link_class, area_type, period)), m AS (SELECT link_class, area_type, period, MEDIAN(ABS(dev)) AS mad FROM d GROUP BY 1, 2, 3), scored AS (SELECT d.*, m.mad, CASE WHEN d.n >= 20 AND m.mad > 0 THEN d.dev / (1.4826 * m.mad) END AS zscore, (d.n >= 20 AND d.capacity_vph > 0 AND d.med_vol > 0 AND d.volume < 0.02 * d.med_vol) AS too_low FROM d JOIN m USING (link_class, area_type, period)) SELECT * FROM scored WHERE vc_ratio >= 0.95 OR (ABS(zscore) >= 3.0 AND COALESCE(vc_ratio, 0) >= 0.25) OR too_low ORDER BY vc_ratio DESC NULLS LAST, ABS(zscore) DESC NULLS LAST

[src: findings/65aa8257d2d8; link_flows.csv (link_flows) row 3093 col volume]

Likely cause: Demand concentrated on this link because parallel routes are missing or slow, or the capacity/lanes are under-coded.

[src: findings/65aa8257d2d8; link_flows.csv (link_flows) row 3093 col volume]

Suggested action: Check lanes and capacity coding on this link and its competitors; compare with counts if available.

[src: findings/65aa8257d2d8; link_flows.csv (link_flows) row 3093 col volume]

Evidence

Metric	Value	Threshold
issue	over_capacity	
volume	26,280	
vc_ratio	1.46	
zscore	9.417	
group_median_volume	5,160	
group_size	40	
link_class	FWY	
area_type	URB	
capacity_vph	6,000	
lanes	3	
too_low	no	
vc_critical		1.30
vc_high		1.10
vc_medium		0.95
zscore_threshold		3

Metric	Value	Threshold
min_group_size		20
low_volume_share_of_group_median		2.0%
min_vc_for_outlier		0.25

[src: findings/65aa8257d2d8; link_flows.csv (link_flows) row 3093 col volume]

Sources: link_flows.csv (link_flows) row 3093 col volume

[src: findings/65aa8257d2d8; link_flows.csv (link_flows) row 3093 col volume]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Add one lane	Add a lane to this link, raising capacity from 6,000 to 8,000 veh/h (one lane = 2,000 veh/h).	V/C falls from 1.46 to about 1.09 if demand is unchanged	sketch elasticity	medium
Verify capacity coding	Check that lanes and capacity per lane match the link's real cross-section; an under-coded capacity produces the same symptom as real congestion.	Not estimated	not computable	medium
Check for missing parallel routes	Confirm that parallel roads in the corridor are coded and open in this scenario; a closed or missing alternative concentrates demand here.	Not estimated	not computable	low

[src: findings/65aa8257d2d8; link_flows.csv (link_flows) row 3093 col volume]

Link volume outliers: Link 197 (FWY, 1106-1107), AM

Critical: Link 197 (FWY) carries 27,720 vehicles in the AM peak, 1.54 times its capacity.

Modeller view: Method: V/C from link_flows; robust $z = (\ln(v+1) - \text{median}) / (1.4826 \text{ MAD})$ within link_class x area_type x period

Values: issue=over_capacity, volume=2.772e+04, vc_ratio=1.54, zscore=8.399, group_median_volume=6796, group_size=24, link_class=FWY, area_type=CBD, capacity_vph=6000, lanes=3, too_low=False

Thresholds: vc_critical=1.3, vc_high=1.1, vc_medium=0.95, zscore_threshold=3, min_group_size=20, low_volume_share_of_group_median=0.02, min_vc_for_outlier=0.25

Files: link_flows.csv rows 3101

SQL: WITH ph(period, hours) AS (VALUES ('AM', 3.0), ('PM', 3.0)), f AS (SELECT f.link_id, f.period, f.volume, COALESCE(f.vc_ratio, f.volume / NULLIF(l.capacity_vph * ph.hours, 0)) AS vc_ratio, l.link_class, l.area_type, l.capacity_vph, l.lanes, f.source_file, f.source_row FROM link_flows f JOIN links l USING (link_id) JOIN ph ON ph.period = f.period WHERE f.user_class = 'ALL' AND f.volume IS NOT NULL AND l.link_class NOT IN ('CONN')), g AS (SELECT link_class, area_type, period, COUNT(*) AS n, MEDIAN(LN(volume + 1)) AS med_log, MEDIAN(volume) AS med_vol FROM f GROUP BY 1, 2, 3), d AS (SELECT f.*, g.n, g.med_log, g.med_vol, LN(f.volume + 1) - g.med_log AS dev FROM f JOIN g USING (link_class, area_type, period)), m AS (SELECT link_class, area_type, period, MEDIAN(ABS(dev)) AS mad FROM d GROUP BY 1, 2, 3), scored AS (SELECT d.*, m.mad, CASE WHEN d.n >= 20 AND m.mad > 0 THEN d.dev / (1.4826 * m.mad) END AS zscore, (d.n >= 20 AND d.capacity_vph > 0 AND d.med_vol > 0 AND d.volume < 0.02 * d.med_vol) AS too_low FROM d JOIN m USING (link_class, area_type, period)) SELECT * FROM scored WHERE vc_ratio >= 0.95 OR (ABS(zscore) >= 3.0 AND COALESCE(vc_ratio, 0) >= 0.25) OR too_low ORDER BY vc_ratio DESC NULLS LAST, ABS(zscore) DESC NULLS LAST

[src: findings/81d9e99d5ab0; link_flows.csv (link_flows) row 3101 col volume]

Likely cause: Demand concentrated on this link because parallel routes are missing or slow, or the capacity/lanes are under-coded.

[src: findings/81d9e99d5ab0; link_flows.csv (link_flows) row 3101 col volume]

Suggested action: Check lanes and capacity coding on this link and its competitors; compare with counts if available.

[src: findings/81d9e99d5ab0; link_flows.csv (link_flows) row 3101 col volume]

Evidence

Metric	Value	Threshold
issue	over_capacity	
volume	27,720	
vc_ratio	1.54	
zscore	8.399	
group_median_volume	6,800	
group_size	24	
link_class	FWY	
area_type	CBD	
capacity_vph	6,000	
lanes	3	
too_low	no	
vc_critical		1.30
vc_high		1.10
vc_medium		0.95
zscore_threshold		3
min_group_size		20
low_volume_share_of_group_median		2.0%
min_vc_for_outlier		0.25

[src: findings/81d9e99d5ab0; link_flows.csv (link_flows) row 3101 col volume]

Sources: link_flows.csv (link_flows) row 3101 col volume

[src: findings/81d9e99d5ab0; link_flows.csv (link_flows) row 3101 col volume]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Add one lane	Add a lane to this link, raising capacity from 6,000 to 8,000 veh/h (one lane = 2,000 veh/h).	V/C falls from 1.54 to about 1.16 if demand is unchanged	sketch elasticity	medium
Verify capacity coding	Check that lanes and capacity per lane match the link's real cross-section; an under-coded capacity produces the same symptom as real congestion.	Not estimated	not computable	medium
Check for missing parallel routes	Confirm that parallel roads in the corridor are coded and open in this scenario; a closed or missing alternative concentrates demand here.	Not estimated	not computable	low

[src: findings/81d9e99d5ab0; link_flows.csv (link_flows) row 3101 col volume]

Land-use control totals: POP / ALL

Critical: Total population across all zones is 283,630, 10.7% below the control total of 317,666.

Modeller view: Method: SUM(land_use.value) per variable x region vs control_totals

Values: variable=POP, region=ALL, modelled=2.836e+05, control=3.177e+05, abs_diff_share=0.1071, scale_factor=1.12, land_use_rows=60

Thresholds: tolerance_share=0.02, severity_bands=['abs_diff_share > 0.10', 'abs_diff_share > 0.05', 'abs_diff_share > 0.02']

Files: control_totals.csv rows 1

SQL: WITH by_region AS (SELECT l.variable, COALESCE(z.region, 'UNKNOWN') AS region, SUM(l.value) AS modelled, COUNT(*) AS n_rows FROM land_use l LEFT JOIN zones z USING (zone_id) GROUP BY 1, 2), all_regions AS (SELECT variable, 'ALL' AS region, SUM(value) AS modelled,

COUNT(*) AS n_rows FROM land_use GROUP BY 1), modelled AS (SELECT * FROM by_region UNION ALL SELECT * FROM all_regions) SELECT c.variable, c.region, c.value AS control, COALESCE(m.modelled, 0) AS modelled, COALESCE(m.n_rows, 0) AS n_rows, c.source_file, c.source_row, ABS(COALESCE(m.modelled, 0) - c.value) / NULLIF(ABS(c.value), 0) AS abs_diff_share FROM control_totals c LEFT JOIN modelled m USING (variable, region) ORDER BY c.variable, c.region

[src: findings/8e85b004d345; control_totals.csv (control_totals) row 1 col value]

Likely cause: Zonal land-use file not rescaled to the latest control totals, or zones missing/duplicated in the land-use file.

[src: findings/8e85b004d345; control_totals.csv (control_totals) row 1 col value]

Suggested action: Scale POP across all zones by 1.120 to match the control total, or confirm the control total is current.

[src: findings/8e85b004d345; control_totals.csv (control_totals) row 1 col value]

Evidence

Metric	Value	Threshold
variable	POP	
region	ALL	
modelled	283,630	
control	317,666	
abs_diff_share	10.7%	
scale_factor	1.12	
land_use_rows	60	
tolerance_share		2.0%
severity_bands		abs_diff_share > 0.10, abs_diff_share > 0.05, abs_diff_share > 0.02

[src: findings/8e85b004d345; control_totals.csv (control_totals) row 1 col value]

Sources: control_totals.csv (control_totals) row 1 col value

[src: findings/8e85b004d345; control_totals.csv (control_totals) row 1 col value]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Scale POP by 1.120	Multiply POP in region ALL by 1.120 so the zonal total (283,630) matches the control total (317,666).	total becomes 317,666; trip ends move roughly in proportion	sketch elasticity	high
Confirm the control total	Check that the control total is the approved figure for this horizon year.	Not estimated	not computable	low

[src: findings/8e85b004d345; control_totals.csv (control_totals) row 1 col value]

Link volume outliers: Link 185 (FWY, 1100-1101), AM

Critical: Link 185 (FWY) carries 25,560 vehicles in the AM peak, 1.42 times its capacity.

Modeller view: Method: V/C from link_flows; robust $z = (\ln(v+1) - \text{median}) / (1.4826 \text{ MAD})$ within link_class x area_type x period

Values: issue=over_capacity, volume=2.556e+04, vc_ratio=1.42, zscore=11.96, group_median_volume=4248, group_size=84, link_class=FWY, area_type=SUB, capacity_vph=6000, lanes=3, too_low=False

Thresholds: vc_critical=1.3, vc_high=1.1, vc_medium=0.95, zscore_threshold=3, min_group_size=20, low_volume_share_of_group_median=0.02, min_vc_for_outlier=0.25

Files: link_flows.csv rows 3089

SQL: WITH ph(period, hours) AS (VALUES ('AM', 3.0), ('PM', 3.0)), f AS (SELECT f.link_id, f.period, f.volume, COALESCE(f.vc_ratio, f.volume / NULLIF(l.capacity_vph * ph.hours, 0)) AS vc_ratio, l.link_class, l.area_type, l.capacity_vph, l.lanes, f.source_file, f.source_row FROM link_flows f JOIN links l USING (link_id) JOIN ph ON ph.period = f.period WHERE f.user_class = 'ALL' AND f.volume IS NOT NULL AND l.link_class NOT IN ('CONN')), g AS (SELECT link_class, area_type, period, COUNT(*) AS n, MEDIAN(LN(volume + 1)) AS med_log, MEDIAN(volume) AS med_vol FROM f GROUP BY 1, 2, 3), d AS (SELECT f.*, g.n, g.med_log, g.med_vol, LN(f.volume + 1) - g.med_log AS dev FROM f JOIN g USING (link_class, area_type, period)), m AS (SELECT link_class, area_type, period, MEDIAN(ABS(dev)) AS mad FROM d GROUP BY 1, 2, 3), scored AS (SELECT d.*, m.mad, CASE WHEN d.n >= 20 AND m.mad > 0 THEN d.dev / (1.4826 * m.mad) END AS zscore, (d.n >= 20 AND d.capacity_vph > 0 AND d.med_vol > 0 AND d.volume < 0.02 * d.med_vol) AS too_low FROM d JOIN m USING (link_class, area_type, period)) SELECT * FROM scored WHERE vc_ratio >= 0.95 OR (ABS(zscore) >= 3.0 AND COALESCE(vc_ratio, 0) >= 0.25) OR too_low ORDER BY vc_ratio DESC NULLS LAST, ABS(zscore) DESC NULLS LAST

[src: findings/95aff43d2c28; link_flows.csv (link_flows) row 3089 col volume]

Likely cause: Demand concentrated on this link because parallel routes are missing or slow, or the capacity/lanes are under-coded.

[src: findings/95aff43d2c28; link_flows.csv (link_flows) row 3089 col volume]

Suggested action: Check lanes and capacity coding on this link and its competitors; compare with counts if available.

[src: findings/95aff43d2c28; link_flows.csv (link_flows) row 3089 col volume]

Evidence

Metric	Value	Threshold
issue	over_capacity	
volume	25,560	
vc_ratio	1.42	
zscore	12.0	
group_median_volume	4,250	
group_size	84	
link_class	FWY	
area_type	SUB	
capacity_vph	6,000	
lanes	3	
too_low	no	
vc_critical		1.30
vc_high		1.10
vc_medium		0.95
zscore_threshold		3
min_group_size		20
low_volume_share_of_group_median		2.0%
min_vc_for_outlier		0.25

[src: findings/95aff43d2c28; link_flows.csv (link_flows) row 3089 col volume]

Sources: link_flows.csv (link_flows) row 3089 col volume

[src: findings/95aff43d2c28; link_flows.csv (link_flows) row 3089 col volume]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Add one lane	Add a lane to this link, raising capacity from 6,000 to 8,000 veh/h (one lane = 2,000 veh/h).	V/C falls from 1.42 to about 1.06 if demand is unchanged	sketch elasticity	medium
Verify capacity coding	Check that lanes and capacity per lane match the link's real cross-section; an under-coded capacity produces the same symptom as real congestion.	Not estimated	not computable	medium
Check for missing parallel routes	Confirm that parallel roads in the corridor are coded and open in this scenario; a closed or missing alternative concentrates demand here.	Not estimated	not computable	low

[src: findings/95aff43d2c28; link_flows.csv (link_flows) row 3089 col volume]

Link volume outliers: Link 201 (FWY, 1108-1109), AM

Critical: Link 201 (FWY) carries 28,440 vehicles in the AM peak, 1.58 times its capacity.

Modeller view: Method: V/C from link_flows; robust $z = (\ln(v+1) - \text{median}) / (1.4826 \text{ MAD})$ within link_class x area_type x period

Values: issue=over_capacity, volume=2.844e+04, vc_ratio=1.58, zscore=9.874, group_median_volume=5162, group_size=40, link_class=FWY, area_type=URB, capacity_vph=6000, lanes=3, too_low=False

Thresholds: vc_critical=1.3, vc_high=1.1, vc_medium=0.95, zscore_threshold=3, min_group_size=20, low_volume_share_of_group_median=0.02, min_vc_for_outlier=0.25

Files: link_flows.csv rows 3105

SQL: WITH ph(period, hours) AS (VALUES ('AM', 3.0), ('PM', 3.0)), f AS (SELECT f.link_id, f.period, f.volume, COALESCE(f.vc_ratio, f.volume / NULLIF(l.capacity_vph * ph.hours, 0)) AS vc_ratio, l.link_class, l.area_type, l.capacity_vph, l.lanes, f.source_file, f.source_row FROM link_flows f JOIN links l USING (link_id) JOIN ph ON ph.period = f.period WHERE f.user_class = 'ALL' AND f.volume IS NOT NULL AND l.link_class NOT IN ('CONN')), g AS (SELECT link_class, area_type, period, COUNT(*) AS n, MEDIAN(LN(volume + 1)) AS med_log, MEDIAN(volume) AS med_vol FROM f GROUP BY 1, 2, 3), d AS (SELECT f.*, g.n, g.med_log, g.med_vol, LN(f.volume + 1) - g.med_log AS dev FROM f JOIN g USING (link_class, area_type, period)), m AS (SELECT link_class, area_type, period, MEDIAN(ABS(dev)) AS mad FROM d GROUP BY 1, 2, 3), scored AS (SELECT d.*, m.mad, CASE WHEN d.n >= 20 AND m.mad > 0 THEN d.dev / (1.4826 * m.mad) END AS zscore, (d.n >= 20 AND d.capacity_vph > 0 AND d.med_vol > 0 AND d.volume < 0.02 * d.med_vol) AS too_low FROM d JOIN m USING (link_class, area_type, period)) SELECT * FROM scored WHERE vc_ratio >= 0.95 OR (ABS(zscore) >= 3.0 AND COALESCE(vc_ratio, 0) >= 0.25) OR too_low ORDER BY vc_ratio DESC NULLS LAST, ABS(zscore) DESC NULLS LAST

[src: findings/9914fb4f0d20; link_flows.csv (link_flows) row 3105 col volume]

Likely cause: Demand concentrated on this link because parallel routes are missing or slow, or the capacity/lanes are under-coded.

[src: findings/9914fb4f0d20; link_flows.csv (link_flows) row 3105 col volume]

Suggested action: Check lanes and capacity coding on this link and its competitors; compare with counts if available.

[src: findings/9914fb4f0d20; link_flows.csv (link_flows) row 3105 col volume]

Evidence

Metric	Value	Threshold
issue	over_capacity	
volume	28,440	
vc_ratio	1.58	
zscore	9.874	
group_median_volume	5,160	

Metric	Value	Threshold
group_size	40	
link_class	FWY	
area_type	URB	
capacity_vph	6,000	
lanes	3	
too_low	no	
vc_critical		1.30
vc_high		1.10
vc_medium		0.95
zscore_threshold		3
min_group_size		20
low_volume_share_of_group_median		2.0%
min_vc_for_outlier		0.25

[src: findings/9914fb4f0d20; link_flows.csv (link_flows) row 3105 col volume]

Sources: link_flows.csv (link_flows) row 3105 col volume

[src: findings/9914fb4f0d20; link_flows.csv (link_flows) row 3105 col volume]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Add one lane	Add a lane to this link, raising capacity from 6,000 to 8,000 veh/h (one lane = 2,000 veh/h).	V/C falls from 1.58 to about 1.19 if demand is unchanged	sketch elasticity	medium
Verify capacity coding	Check that lanes and capacity per lane match the link's real cross-section; an under-coded capacity produces the same symptom as real congestion.	Not estimated	not computable	medium
Check for missing parallel routes	Confirm that parallel roads in the corridor are coded and open in this scenario; a closed or missing alternative concentrates demand here.	Not estimated	not computable	low

[src: findings/9914fb4f0d20; link_flows.csv (link_flows) row 3105 col volume]

Network connectivity: Link 6 (ART, 1004-1003)

Critical: Link 6 (ART) has a capacity of 0 vehicles per hour, so traffic cannot realistically use it.

Modeller view: Method: attribute threshold on links

Values: issue=zero_capacity, capacity_vph=0, ffs_kph=60, lanes=2, link_class=ART

Thresholds: min_capacity_vph=1, min_ffs_kph=5

Files: links.csv rows 6

SQL: SELECT link_id, link_class, capacity_vph, ffs_kph, lanes, source_file, source_row FROM links WHERE capacity_vph < 1.0 OR ffs_kph < 5.0 ORDER BY link_id

[src: findings/ad89fdca2740; links.csv (links) row 6 col capacity_vph]

Likely cause: Capacity or speed lookup missed this link class/area type, or the link was meant to be closed.

[src: findings/ad89fdca2740; links.csv (links) row 6 col capacity_vph]

Suggested action: Fix the link attributes or remove the link from the scenario network.

[src: findings/ad89fdca2740; links.csv (links) row 6 col capacity_vph]

Evidence

Metric	Value	Threshold
issue	zero_capacity	

Metric	Value	Threshold
capacity_vph	0	
ffs_kph	60	
lanes	2	
link_class	ART	
min_capacity_vph		0
min_ffs_kph		5

[src: findings/ad89fdca2740; links.csv (links) row 6 col capacity_vph]

Sources: links.csv (links) row 6 col capacity_vph

[src: findings/ad89fdca2740; links.csv (links) row 6 col capacity_vph]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Repair the network coding	Fix the attribute or add the missing link/one-way direction and re-assign.	Not estimated	not computable	low

[src: findings/ad89fdca2740; links.csv (links) row 6 col capacity_vph]

Link volume outliers: Link 183 (FWY, 1099-1100), AM

Critical: Link 183 (FWY) carries 25,200 vehicles in the AM peak, 1.40 times its capacity.

Modeller view: Method: V/C from link_flows; robust $z = (\ln(v+1) - \text{median}) / (1.4826 \text{ MAD})$ within link_class x area_type x period

Values: issue=over_capacity, volume=2.52e+04, vc_ratio=1.4, zscore=11.86, group_median_volume=4248, group_size=84, link_class=FWY, area_type=SUB, capacity_vph=6000, lanes=3, too_low=False

Thresholds: vc_critical=1.3, vc_high=1.1, vc_medium=0.95, zscore_threshold=3, min_group_size=20, low_volume_share_of_group_median=0.02, min_vc_for_outlier=0.25

Files: link_flows.csv rows 3087

SQL: WITH ph(period, hours) AS (VALUES ('AM', 3.0), ('PM', 3.0)), f AS (SELECT f.link_id, f.period, f.volume, COALESCE(f.vc_ratio, f.volume / NULLIF(l.capacity_vph * ph.hours, 0)) AS vc_ratio, l.link_class, l.area_type, l.capacity_vph, l.lanes, f.source_file, f.source_row FROM link_flows f JOIN links l USING (link_id) JOIN ph ON ph.period = f.period WHERE f.user_class = 'ALL' AND f.volume IS NOT NULL AND l.link_class NOT IN ('CONN')), g AS (SELECT link_class, area_type, period, COUNT(*) AS n, MEDIAN(LN(volume + 1)) AS med_log, MEDIAN(volume) AS med_vol FROM f GROUP BY 1, 2, 3), d AS (SELECT f.*, g.n, g.med_log, g.med_vol, LN(f.volume + 1) - g.med_log AS dev FROM f JOIN g USING (link_class, area_type, period)), m AS (SELECT link_class, area_type, period, MEDIAN(ABS(dev)) AS mad FROM d GROUP BY 1, 2, 3), scored AS (SELECT d.*, m.mad, CASE WHEN d.n >= 20 AND m.mad > 0 THEN d.dev / (1.4826 * m.mad) END AS zscore, (d.n >= 20 AND d.capacity_vph > 0 AND d.med_vol > 0 AND d.volume < 0.02 * d.med_vol) AS too_low FROM d JOIN m USING (link_class, area_type, period)) SELECT * FROM scored WHERE vc_ratio >= 0.95 OR (ABS(zscore) >= 3.0 AND COALESCE(vc_ratio, 0) >= 0.25) OR too_low ORDER BY vc_ratio DESC NULLS LAST, ABS(zscore) DESC NULLS LAST

[src: findings/baafae99d76e; link_flows.csv (link_flows) row 3087 col volume]

Likely cause: Demand concentrated on this link because parallel routes are missing or slow, or the capacity/lanes are under-coded.

[src: findings/baafae99d76e; link_flows.csv (link_flows) row 3087 col volume]

Suggested action: Check lanes and capacity coding on this link and its competitors; compare with counts if available.

[src: findings/baafae99d76e; link_flows.csv (link_flows) row 3087 col volume]

Evidence

Metric	Value	Threshold
issue	over_capacity	
volume	25,200	
vc_ratio	1.40	
zscore	11.9	
group_median_volume	4,250	
group_size	84	
link_class	FWY	
area_type	SUB	
capacity_vph	6,000	
lanes	3	
too_low	no	
vc_critical		1.30
vc_high		1.10
vc_medium		0.95
zscore_threshold		3
min_group_size		20
low_volume_share_of_group_median		2.0%
min_vc_for_outlier		0.25

[src: findings/baafae99d76e; link_flows.csv (link_flows) row 3087 col volume]

Sources: link_flows.csv (link_flows) row 3087 col volume

[src: findings/baafae99d76e; link_flows.csv (link_flows) row 3087 col volume]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Add one lane	Add a lane to this link, raising capacity from 6,000 to 8,000 veh/h (one lane = 2,000 veh/h).	V/C falls from 1.40 to about 1.05 if demand is unchanged	sketch elasticity	medium
Verify capacity coding	Check that lanes and capacity per lane match the link's real cross-section; an under-coded capacity produces the same symptom as real congestion.	Not estimated	not computable	medium
Check for missing parallel routes	Confirm that parallel roads in the corridor are coded and open in this scenario; a closed or missing alternative concentrates demand here.	Not estimated	not computable	low

[src: findings/baafae99d76e; link_flows.csv (link_flows) row 3087 col volume]

Convergence and noise band: HWY_ASSIGN AM

Critical: The highway assignment for the AM peak stopped at a relative gap of 2.00% after 20 iterations, above the 0.10% target.

Modeller view: Method: REL_GAP at the last iteration per stage x period

Values: stage=HWY_ASSIGN, rel_gap=0.02, iterations=20

Thresholds: rel_gap_target=0.001, rel_gap_high=0.01

Files: convergence.csv rows 58

SQL: SELECT stage, period, iteration, value AS rel_gap, source_file, source_row FROM convergence c WHERE metric = 'REL_GAP' AND stage IN ('HWY_ASSIGN', 'PT_ASSIGN') AND iteration = (SELECT MAX(iteration) FROM convergence x WHERE x.metric = 'REL_GAP' AND x.stage = c.stage AND COALESCE(x.period, '') = COALESCE(c.period, '')) ORDER BY stage, period

[src: findings/db2b60bb79ca; convergence.csv (convergence) row 58 col value]

Likely cause: Too few iterations, or a network coding problem (very high V/C, unrealistic capacities) that prevents equilibrium.

[src: findings/db2b60bb79ca; convergence.csv (convergence) row 58 col value]

Suggested action: Increase the iteration limit or tighten the gap criterion, and resolve any capacity findings first.

[src: findings/db2b60bb79ca; convergence.csv (convergence) row 58 col value]

Evidence

Metric	Value	Threshold
stage	HWY_ASSIGN	
rel_gap	0.02	
iterations	20	
rel_gap_target		0.001
rel_gap_high		0.01

[src: findings/db2b60bb79ca; convergence.csv (convergence) row 58 col value]

Sources: convergence.csv (convergence) row 58 col value

[src: findings/db2b60bb79ca; convergence.csv (convergence) row 58 col value]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Increase the iteration limit	Raise the maximum iterations and tighten the gap criterion for this period; re-run before interpreting link-level differences.	Not estimated	not computable	low
Fix capacity findings first	Links with V/C far above 1 slow or prevent convergence.	Not estimated	not computable	low

[src: findings/db2b60bb79ca; convergence.csv (convergence) row 58 col value]

Link volume outliers: Link 187 (FWY, 1101-1102), AM

Critical: Link 187 (FWY) carries 25,920 vehicles in the AM peak, 1.44 times its capacity.

Modeller view: Method: V/C from link_flows; robust $z = (\ln(v+1) - \text{median}) / (1.4826 \text{ MAD})$ within link_class x area_type x period

Values: issue=over_capacity, volume=2.592e+04, vc_ratio=1.44, zscore=9.337, group_median_volume=5162, group_size=40, link_class=FWY, area_type=URB, capacity_vph=6000, lanes=3, too_low=False

Thresholds: vc_critical=1.3, vc_high=1.1, vc_medium=0.95, zscore_threshold=3, min_group_size=20, low_volume_share_of_group_median=0.02, min_vc_for_outlier=0.25

Files: link_flows.csv rows 3091

SQL: WITH ph(period, hours) AS (VALUES ('AM', 3.0), ('PM', 3.0)), f AS (SELECT f.link_id, f.period, f.volume, COALESCE(f.vc_ratio, f.volume / NULLIF(l.capacity_vph * ph.hours, 0)) AS vc_ratio, l.link_class, l.area_type, l.capacity_vph, l.lanes, f.source_file, f.source_row FROM link_flows f JOIN links l USING (link_id) JOIN ph ON ph.period = f.period WHERE f.user_class = 'ALL' AND f.volume IS NOT NULL AND l.link_class NOT IN ('CONN')), g AS (SELECT link_class, area_type, period, COUNT(*) AS n, MEDIAN(LN(volume + 1)) AS med_log, MEDIAN(volume) AS med_vol FROM f GROUP BY 1, 2, 3), d AS (SELECT f.*, g.n, g.med_log, g.med_vol, LN(f.volume + 1) - g.med_log AS dev FROM f JOIN g USING (link_class, area_type, period)), m AS (SELECT link_class, area_type, period, MEDIAN(ABS(dev)) AS mad FROM d GROUP BY 1, 2, 3), scored AS (SELECT d.*, m.mad, CASE WHEN d.n >= 20 AND m.mad > 0 THEN d.dev / (1.4826 * m.mad) END AS zscore, (d.n >= 20 AND d.capacity_vph > 0 AND d.med_vol > 0 AND d.volume < 0.02 * d.med_vol) AS too_low FROM d JOIN m USING (link_class, area_type, period)) SELECT * FROM scored WHERE vc_ratio >= 0.95 OR (ABS(zscore) >= 3.0 AND COALESCE(vc_ratio, 0) >= 0.25) OR too_low ORDER BY vc_ratio DESC NULLS LAST, ABS(zscore) DESC NULLS LAST

[src: findings/db859fde7a54; link_flows.csv (link_flows) row 3091 col volume]

Likely cause: Demand concentrated on this link because parallel routes are missing or slow, or the capacity/lanes are under-coded.

[src: findings/db859fde7a54; link_flows.csv (link_flows) row 3091 col volume]

Suggested action: Check lanes and capacity coding on this link and its competitors; compare with counts if available.

[src: findings/db859fde7a54; link_flows.csv (link_flows) row 3091 col volume]

Evidence

Metric	Value	Threshold
issue	over_capacity	
volume	25,920	
vc_ratio	1.44	
zscore	9.337	
group_median_volume	5,160	
group_size	40	
link_class	FWY	
area_type	URB	
capacity_vph	6,000	
lanes	3	
too_low	no	
vc_critical		1.30
vc_high		1.10
vc_medium		0.95
zscore_threshold		3
min_group_size		20
low_volume_share_of_group_median		2.0%
min_vc_for_outlier		0.25

[src: findings/db859fde7a54; link_flows.csv (link_flows) row 3091 col volume]

Sources: link_flows.csv (link_flows) row 3091 col volume

[src: findings/db859fde7a54; link_flows.csv (link_flows) row 3091 col volume]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Add one lane	Add a lane to this link, raising capacity from 6,000 to 8,000 veh/h (one lane = 2,000 veh/h).	V/C falls from 1.44 to about 1.08 if demand is unchanged	sketch elasticity	medium
Verify capacity coding	Check that lanes and capacity per lane match the link's real cross-section; an under-coded capacity produces the same symptom as real congestion.	Not estimated	not computable	medium
Check for missing parallel routes	Confirm that parallel roads in the corridor are coded and open in this scenario; a closed or missing alternative concentrates demand here.	Not estimated	not computable	low

[src: findings/db859fde7a54; link_flows.csv (link_flows) row 3091 col volume]

Link volume outliers: Link 195 (FWY, 1105-1106), AM

Critical: Link 195 (FWY) carries 27,360 vehicles in the AM peak, 1.52 times its capacity.

Modeller view: Method: V/C from link_flows; robust $z = (\ln(v+1) - \text{median}) / (1.4826 \text{ MAD})$ within link_class x area_type x period

Values: issue=over_capacity, volume=2.736e+04, vc_ratio=1.52, zscore=8.321,

group_median_volume=6796, group_size=24, link_class=FWY, area_type=CBD, capacity_vph=6000, lanes=3, too_low=False

Thresholds: vc_critical=1.3, vc_high=1.1, vc_medium=0.95, zscore_threshold=3, min_group_size=20, low_volume_share_of_group_median=0.02, min_vc_for_outlier=0.25

Files: link_flows.csv rows 3099

SQL: WITH ph(period, hours) AS (VALUES ('AM', 3.0), ('PM', 3.0)), f AS (SELECT f.link_id, f.period, f.volume, COALESCE(f.vc_ratio, f.volume / NULLIF(l.capacity_vph * ph.hours, 0)) AS vc_ratio, l.link_class, l.area_type, l.capacity_vph, l.lanes, f.source_file, f.source_row FROM link_flows f JOIN links l USING (link_id) JOIN ph ON ph.period = f.period WHERE f.user_class = 'ALL' AND f.volume IS NOT NULL AND l.link_class NOT IN ('CONN')), g AS (SELECT link_class, area_type, period, COUNT(*) AS n, MEDIAN(LN(volume + 1)) AS med_log, MEDIAN(volume) AS med_vol FROM f GROUP BY 1, 2, 3), d AS (SELECT f.*, g.n, g.med_log, g.med_vol, LN(f.volume + 1) - g.med_log AS dev FROM f JOIN g USING (link_class, area_type, period)), m AS (SELECT link_class, area_type, period, MEDIAN(ABS(dev)) AS mad FROM d GROUP BY 1, 2, 3), scored AS (SELECT d.*, m.mad, CASE WHEN d.n >= 20 AND m.mad > 0 THEN d.dev / (1.4826 * m.mad) END AS zscore, (d.n >= 20 AND d.capacity_vph > 0 AND d.med_vol > 0 AND d.volume < 0.02 * d.med_vol) AS too_low FROM d JOIN m USING (link_class, area_type, period)) SELECT * FROM scored WHERE vc_ratio >= 0.95 OR (ABS(zscore) >= 3.0 AND COALESCE(vc_ratio, 0) >= 0.25) OR too_low ORDER BY vc_ratio DESC NULLS LAST, ABS(zscore) DESC NULLS LAST

[src: findings/de5af663021e; link_flows.csv (link_flows) row 3099 col volume]

Likely cause: Demand concentrated on this link because parallel routes are missing or slow, or the capacity/lanes are under-coded.

[src: findings/de5af663021e; link_flows.csv (link_flows) row 3099 col volume]

Suggested action: Check lanes and capacity coding on this link and its competitors; compare with counts if available.

[src: findings/de5af663021e; link_flows.csv (link_flows) row 3099 col volume]

Evidence

Metric	Value	Threshold
issue	over_capacity	
volume	27,360	
vc_ratio	1.52	
zscore	8.321	
group_median_volume	6,800	
group_size	24	
link_class	FWY	
area_type	CBD	
capacity_vph	6,000	
lanes	3	
too_low	no	
vc_critical		1.30
vc_high		1.10
vc_medium		0.95
zscore_threshold		3
min_group_size		20
low_volume_share_of_group_median		2.0%
min_vc_for_outlier		0.25

[src: findings/de5af663021e; link_flows.csv (link_flows) row 3099 col volume]

Sources: link_flows.csv (link_flows) row 3099 col volume

[src: findings/de5af663021e; link_flows.csv (link_flows) row 3099 col volume]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Add one lane	Add a lane to this link, raising capacity from 6,000 to 8,000 veh/h (one lane = 2,000 veh/h).	V/C falls from 1.52 to about 1.14 if demand is unchanged	sketch elasticity	medium
Verify capacity coding	Check that lanes and capacity per lane match the link's real cross-section; an under-coded capacity produces the same symptom as real congestion.	Not estimated	not computable	medium
Check for missing parallel routes	Confirm that parallel roads in the corridor are coded and open in this scenario; a closed or missing alternative concentrates demand here.	Not estimated	not computable	low

[src: findings/de5af663021e; link_flows.csv (link_flows) row 3099 col volume]

High findings (13)

Unrealistic speeds and times: Link 28 (LOC, 1016-1015), AM

High: Traffic on link 28 (LOC) crawls at 2.0 km/h in the AM peak.

Modeller view: Method: link_flows joined to links; thresholds on speed ratio, floor, delay

Values: issue=speed_below_floor, cong_speed_kph=2, ffs_kph=40, delay_s=5.01, volume=36, speed_ratio=0.05, link_class=LOC

Thresholds: min_cong_speed_kph=5, max_speed_over_ffs_ratio=1.05, max_delay_s=900

Files: link_flows.csv rows 2932

SQL: SELECT f.link_id, f.period, f.volume, f.cong_speed_kph, f.delay_s, f.cong_time_s, l.ffs_kph, l.link_class, l.length_m, f.source_file, f.source_row, f.cong_speed_kph / NULLIF(l.ffs_kph, 0) AS speed_ratio FROM link_flows f JOIN links l USING (link_id) WHERE f.user_class = 'ALL' AND f.period IN ('AM', 'PM') AND ((f.cong_speed_kph IS NOT NULL AND l.ffs_kph > 0 AND f.cong_speed_kph > 1.05 * l.ffs_kph) OR (f.cong_speed_kph IS NOT NULL AND f.cong_speed_kph < 5.0) OR (f.delay_s IS NOT NULL AND f.delay_s > 900.0)) ORDER BY f.link_id, f.period

[src: findings/0dcfc202ff8d; link_flows.csv (link_flows) row 2932 col cong_speed_kph]

Likely cause: Severe over-capacity or a volume-delay function that collapses at high V/C; sometimes a zero-length or zero-speed coding.

[src: findings/0dcfc202ff8d; link_flows.csv (link_flows) row 2932 col cong_speed_kph]

Suggested action: Check V/C on this link, the volume-delay parameters for its class, and the free-flow speed coding.

[src: findings/0dcfc202ff8d; link_flows.csv (link_flows) row 2932 col cong_speed_kph]

Evidence

Metric	Value	Threshold
issue	speed_below_floor	
issues	speed_below_floor	
cong_speed_kph	2 km/h	
ffs_kph	40	
delay_s	5.01	
volume	40	
speed_ratio	0.05	
link_class	LOC	
min_cong_speed_kph		5 km/h
max_speed_over_ffs_ratio		1.05
max_delay_s		900

[src: findings/0dcfc202ff8d; link_flows.csv (link_flows) row 2932 col cong_speed_kph]

Sources: link_flows.csv (link_flows) row 2932 col cong_speed_kph

[src: findings/0dcfc202ff8d; link_flows.csv (link_flows) row 2932 col cong_speed_kph]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Review volume-delay parameters	Check the BPR/conical parameters for this link class; extreme delay or near-zero speed usually means the curve explodes above V/C 1.2.	Not estimated	not computable	low
Resolve capacity on this link first	If a V/C finding exists for the same link, fixing it removes this symptom.	Not estimated	not computable	low

[src: findings/0dcfc202ff8d; link_flows.csv (link_flows) row 2932 col cong_speed_kph]

Parameter drift: Parameter PT_FARE_BASE

High: Parameter PT_FARE_BASE is set to 2.0 but the approved value is 2.5.

Modeller view: Method: parameters.value vs approved_value (numeric or text)

Values: key=PT_FARE_BASE, value=2.0, approved_value=2.5, change_share=-0.2, drifted=True

Thresholds: numeric_tolerance=1e-09

Files: parameters.csv rows 6

SQL: SELECT key, value, approved_value, source_file, source_row, TRY_CAST(value AS DOUBLE) AS v_num, TRY_CAST(approved_value AS DOUBLE) AS a_num FROM parameters WHERE approved_value IS NOT NULL AND CASE WHEN TRY_CAST(value AS DOUBLE) IS NOT NULL AND TRY_CAST(approved_value AS DOUBLE) IS NOT NULL THEN ABS(TRY_CAST(value AS DOUBLE) - TRY_CAST(approved_value AS DOUBLE)) > 1e-09 * GREATEST(1.0, ABS(TRY_CAST(approved_value AS DOUBLE))) ELSE TRIM(COALESCE(value, '')) <> TRIM(approved_value) END ORDER BY key

[src: findings/162b2cc15a9e; parameters.csv (parameters) row 6 col value]

Likely cause: Parameter edited for a test and not reverted, or the approved register is out of date.

[src: findings/162b2cc15a9e; parameters.csv (parameters) row 6 col value]

Suggested action: Restore the approved value, or record the change and get the new value approved.

[src: findings/162b2cc15a9e; parameters.csv (parameters) row 6 col value]

Evidence

Metric	Value	Threshold
key	PT_FARE_BASE	
value	2.0	
approved_value	2.5	
change_share	-20.0%	
drifted	yes	
numeric_tolerance		1e-09

[src: findings/162b2cc15a9e; parameters.csv (parameters) row 6 col value]

Sources: parameters.csv (parameters) row 6 col value

[src: findings/162b2cc15a9e; parameters.csv (parameters) row 6 col value]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Restore the approved value	Reset the parameter to its approved value or get the new value approved.	Not estimated	not computable	low

[src: findings/162b2cc15a9e; parameters.csv (parameters) row 6 col value]

Unused transit services: Line BUS_UNUSED

High: Line BUS_UNUSED (BUS) runs 0.0% full on average across the day, although 12,689 public transport trips start within 1 km of its stops (emptiest in the AM peak at 0.0%).

Modeller view: Method: demand-weighted mean over periods of mean(load)/vehicle_capacity per line; PT productions of zones with centroid within catchment_km of any stop

Values: load_factor=0, vehicle_capacity=80, catchment_demand=1.269e+04, headway_min=10, n_periods_with_demand=5, n_catchment_zones=6, mode=BUS

Thresholds: min_load_factor=0.05, min_catchment_demand=500, catchment_km=1

Files: line_loads.csv rows 105, 456, 222, 573, 339; od.csv

SQL: WITH k AS (SELECT COS(RADIANS(AVG(y))) AS coslat FROM nodes WHERE y IS NOT NULL), cap AS (SELECT line_id, MAX(vehicle_capacity) AS capacity, ANY_VALUE(mode) AS mode FROM transit_lines GROUP BY 1), hw AS (SELECT line_id, period, MAX(headway_min) AS headway_min FROM transit_lines GROUP BY 1, 2), lf AS (SELECT line_id, period, AVG(load) AS mean_load, MAX(load) AS max_load, COUNT(*) AS n_segments, ANY_VALUE(source_file) AS source_file, MIN(source_row) AS source_row FROM line_loads GROUP BY 1, 2), stop_nodes AS (SELECT line_id, from_node AS node_id FROM transit_segments WHERE COALESCE(is_stop, TRUE) UNION SELECT line_id, to_node FROM transit_segments WHERE COALESCE(is_stop, TRUE)), stops AS (SELECT DISTINCT s.line_id, n.x, n.y FROM stop_nodes s JOIN nodes n USING (node_id) WHERE n.x IS NOT NULL AND n.y IS NOT NULL), near AS (SELECT DISTINCT st.line_id, z.zone_id FROM stops st CROSS JOIN k JOIN zones z ON ABS(z.centroid_y - st.y) * 110.57 <= 1.0 AND ABS(z.centroid_x - st.x) * 111.32 * k.coslat <= 1.0 AND POWER((z.centroid_x - st.x) * 111.32 * k.coslat, 2) + POWER((z.centroid_y - st.y) * 110.57, 2) <= 1.0), prod AS (SELECT origin AS zone_id, period, SUM(trips) AS trips FROM od WHERE matrix_kind = 'DEMAND' AND mode IN ('PT') GROUP BY 1, 2), catch AS (SELECT nz.line_id, p.period, SUM(p.trips) AS catchment_demand, COUNT(DISTINCT nz.zone_id) AS n_zones FROM near nz JOIN prod p USING (zone_id) GROUP BY 1, 2) SELECT lf.line_id, lf.period, cap.mode, cap.capacity, hw.headway_min, lf.mean_load, lf.max_load, lf.n_segments, lf.mean_load / NULLIF(cap.capacity, 0) AS load_factor, COALESCE(c.catchment_demand, 0) AS catchment_demand, COALESCE(c.n_zones, 0) AS n_catchment_zones, lf.source_file, lf.source_row FROM lf JOIN cap USING (line_id) LEFT JOIN hw USING (line_id, period) LEFT JOIN catch c USING (line_id, period) ORDER BY lf.line_id, lf.period

[src: findings/28b1ae115a35; line_loads.csv (line_loads) row 105 col load; line_loads.csv (line_loads) row 456 col load; line_loads.csv (line_loads) row 222 col load; line_loads.csv (line_loads) row 573 col load; line_loads.csv (line_loads) row 339 col load; od.csv (od) col trips]

Likely cause: The line is not reachable in the transit network (missing walk access, wrong stop flags, or a broken itinerary), or a parallel service takes all the demand.

[src: findings/28b1ae115a35; line_loads.csv (line_loads) row 105 col load; line_loads.csv (line_loads) row 456 col load; line_loads.csv (line_loads) row 222 col load; line_loads.csv (line_loads) row 573 col load; line_loads.csv (line_loads) row 339 col load; od.csv (od) col trips]

Suggested action: Check stop access links and the line's itinerary; if the service is genuinely unused, reduce its frequency.

[src: findings/28b1ae115a35; line_loads.csv (line_loads) row 105 col load; line_loads.csv (line_loads) row 456 col load; line_loads.csv (line_loads) row 222 col load; line_loads.csv (line_loads) row 573 col load; line_loads.csv (line_loads) row 339 col load; od.csv (od) col trips]

Evidence

Metric	Value	Threshold
load_factor	0	
vehicle_capacity	80	
catchment_demand	12,689	
headway_min	10	
n_periods_with_demand	5	
n_catchment_zones	6	
mode	BUS	

Metric	Value	Threshold
per_period	5 items (full list in findings.json): period=AM, load_factor=0, mean_load=0, ... (6 keys), period=EV, load_factor=0, mean_load=0, ... (6 keys), period=MD, lo...	
min_load_factor		0.05
min_catchment_demand		500
catchment_km		1

[src: findings/28b1ae115a35; line_loads.csv (line_loads) row 105 col load; line_loads.csv (line_loads) row 456 col load; line_loads.csv (line_loads) row 222 col load; line_loads.csv (line_loads) row 573 col load; line_loads.csv (line_loads) row 339 col load; od.csv (od) col trips]

Sources: line_loads.csv (line_loads) row 105 col load; line_loads.csv (line_loads) row 456 col load; line_loads.csv (line_loads) row 222 col load; line_loads.csv (line_loads) row 573 col load; line_loads.csv (line_loads) row 339 col load; od.csv (od) col trips

[src: findings/28b1ae115a35; line_loads.csv (line_loads) row 105 col load; line_loads.csv (line_loads) row 456 col load; line_loads.csv (line_loads) row 222 col load; line_loads.csv (line_loads) row 573 col load; line_loads.csv (line_loads) row 339 col load; od.csv (od) col trips]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Check access to the line	Verify stop flags, walk-access links and the itinerary before changing service: an unreachable line looks identical to an unwanted one.	Not estimated	not computable	low
Halve the frequency	If the service is genuinely unused, double the headway to release vehicles.	saves about 1,100 vehicle-km per day across AM, EV, MD, NT, PM (half the current vehicle trips)	sketch elasticity	low

[src: findings/28b1ae115a35; line_loads.csv (line_loads) row 105 col load; line_loads.csv (line_loads) row 456 col load; line_loads.csv (line_loads) row 222 col load; line_loads.csv (line_loads) row 573 col load; line_loads.csv (line_loads) row 339 col load; od.csv (od) col trips]

Unrealistic speeds and times: Link 27 (LOC, 1015-1016), AM

High: Traffic on link 27 (LOC) crawls at 2.0 km/h in the AM peak.

Modeller view: Method: link_flows joined to links; thresholds on speed ratio, floor, delay

Values: issue=speed_below_floor, cong_speed_kph=2, ffs_kph=40, delay_s=0, volume=36, speed_ratio=0.05, link_class=LOC

Thresholds: min_cong_speed_kph=5, max_speed_over_ffs_ratio=1.05, max_delay_s=900

Files: link_flows.csv rows 2931

SQL: SELECT f.link_id, f.period, f.volume, f.cong_speed_kph, f.delay_s, f.cong_time_s, l.ffs_kph, l.link_class, l.length_m, f.source_file, f.source_row, f.cong_speed_kph / NULLIF(l.ffs_kph, 0) AS speed_ratio FROM link_flows f JOIN links l USING (link_id) WHERE f.user_class = 'ALL' AND f.period IN ('AM', 'PM') AND ((f.cong_speed_kph IS NOT NULL AND l.ffs_kph > 0 AND f.cong_speed_kph > 1.05 * l.ffs_kph) OR (f.cong_speed_kph IS NOT NULL AND f.cong_speed_kph < 5.0) OR (f.delay_s IS NOT NULL AND f.delay_s > 900.0)) ORDER BY f.link_id, f.period

[src: findings/552aedef85c4a; link_flows.csv (link_flows) row 2931 col cong_speed_kph]

Likely cause: Severe over-capacity or a volume-delay function that collapses at high V/C; sometimes a zero-length or zero-speed coding.

[src: findings/552aedef85c4a; link_flows.csv (link_flows) row 2931 col cong_speed_kph]

Suggested action: Check V/C on this link, the volume-delay parameters for its class, and the free-flow speed coding.

[src: findings/552aedef85c4a; link_flows.csv (link_flows) row 2931 col cong_speed_kph]

Evidence

Metric	Value	Threshold
issue	speed_below_floor	
issues	speed_below_floor	
cong_speed_kph	2 km/h	
ffs_kph	40	
delay_s	0	
volume	40	
speed_ratio	0.05	
link_class	LOC	
min_cong_speed_kph		5 km/h
max_speed_over_ffs_ratio		1.05
max_delay_s		900

[src: findings/552aedef85c4a; link_flows.csv (link_flows) row 2931 col cong_speed_kph]

Sources: link_flows.csv (link_flows) row 2931 col cong_speed_kph

[src: findings/552aedef85c4a; link_flows.csv (link_flows) row 2931 col cong_speed_kph]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Review volume-delay parameters	Check the BPR/conical parameters for this link class; extreme delay or near-zero speed usually means the curve explodes above V/C 1.2.	Not estimated	not computable	low
Resolve capacity on this link first	If a V/C finding exists for the same link, fixing it removes this symptom.	Not estimated	not computable	low

[src: findings/552aedef85c4a; link_flows.csv (link_flows) row 2931 col cong_speed_kph]

Centroid connector placement: Link 90003 (CONN, 1100-16)

High: Connector 90003 loads zone traffic straight onto a FWY link at node 1100.

Modeller view: Method: connector street-end node joined to links of high class

Values: issue=connector_on_high_class, street_node=1100, high_class_link=183, high_class=FWY, length_m=9483, centroid=16

Thresholds: high_class_links=['FWY', 'EXP']

Files: links.csv rows 970

SQL: WITH conn AS (SELECT l.link_id, l.a_node, l.b_node, l.length_m, l.link_class, l.source_file, l.source_row, CASE WHEN COALESCE(na.is_centroid, FALSE) THEN l.b_node ELSE l.a_node END AS street_node, CASE WHEN COALESCE(na.is_centroid, FALSE) THEN l.a_node WHEN COALESCE(nb.is_centroid, FALSE) THEN l.b_node END AS centroid FROM links l LEFT JOIN nodes na ON na.node_id = l.a_node LEFT JOIN nodes nb ON nb.node_id = l.b_node WHERE l.link_class IN ('CONN') OR COALESCE(na.is_centroid, FALSE) OR COALESCE(nb.is_centroid, FALSE)), touch AS (SELECT c.link_id, MIN(h.link_id) AS high_link, ANY_VALUE(h.link_class) AS high_class FROM conn c JOIN links h ON (h.a_node = c.street_node OR h.b_node = c.street_node) AND h.link_class IN ('FWY', 'EXP') GROUP BY c.link_id) SELECT c.*, t.high_link, t.high_class FROM conn c LEFT JOIN touch t USING (link_id) WHERE t.high_link IS NOT NULL OR c.length_m > 5000.0 ORDER BY c.link_id

[src: findings/6577bd4f6200; links.csv (links) row 970 col link_id]

Likely cause: Connector snapped to the nearest node, which happens to be on the freeway, rather than to a local street.

[src: findings/6577bd4f6200; links.csv (links) row 970 col link_id]

Suggested action: Move the connector to a local or collector street node near the zone centroid.

[src: findings/6577bd4f6200; links.csv (links) row 970 col link_id]

Evidence

Metric	Value	Threshold
issue	connector_on_high_class	
street_node	1,100	
high_class_link	183	
high_class	FWY	
length_m	9,483	
centroid	16	
high_class_links		FWY, EXP

[src: findings/6577bd4f6200; links.csv (links) row 970 col link_id]

Sources: links.csv (links) row 970 col link_id

[src: findings/6577bd4f6200; links.csv (links) row 970 col link_id]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Relocate the connector to a local street	Move connector 90003 from node 1100 to the nearest collector or local street node so zone traffic enters via ramps/junctions.	removes about 15 vehicles per day (1 veh/h in AM) from the FWY link 183	sketch elasticity	medium
Split the zone	If the zone straddles the freeway, split it so each part connects to its own side of the network.	Not estimated	not computable	low

[src: findings/6577bd4f6200; links.csv (links) row 970 col link_id]

Link volume outliers: Link 588 (COL, 1121-1107), PM

High: Link 588 (COL) carries 2,426 vehicles in the PM peak, 1.16 times its capacity.

Modeller view: Method: V/C from link_flows; robust $z = (\ln(v+1) - \text{median}) / (1.4826 \text{ MAD})$ within link_class x area_type x period

Values: issue=over_capacity, volume=2426, vc_ratio=1.155, zscore=1.415, group_median_volume=851.4, group_size=22, link_class=COL, area_type=CBD, capacity_vph=700, lanes=1, too_low=False

Thresholds: vc_critical=1.3, vc_high=1.1, vc_medium=0.95, zscore_threshold=3, min_group_size=20, low_volume_share_of_group_median=0.02, min_vc_for_outlier=0.25

Files: link_flows.csv rows 11236

SQL: WITH ph(period, hours) AS (VALUES ('AM', 3.0), ('PM', 3.0)), f AS (SELECT f.link_id, f.period, f.volume, COALESCE(f.vc_ratio, f.volume / NULLIF(l.capacity_vph * ph.hours, 0)) AS vc_ratio, l.link_class, l.area_type, l.capacity_vph, l.lanes, f.source_file, f.source_row FROM link_flows f JOIN links l USING (link_id) JOIN ph ON ph.period = f.period WHERE f.user_class = 'ALL' AND f.volume IS NOT NULL AND l.link_class NOT IN ('CONN')), g AS (SELECT link_class, area_type, period, COUNT(*) AS n, MEDIAN(LN(volume + 1)) AS med_log, MEDIAN(volume) AS med_vol FROM f GROUP BY 1, 2, 3), d AS (SELECT f.*, g.n, g.med_log, g.med_vol, LN(f.volume + 1) - g.med_log AS dev FROM f JOIN g USING (link_class, area_type, period)), m AS (SELECT link_class, area_type, period, MEDIAN(ABS(dev)) AS mad FROM d GROUP BY 1, 2, 3), scored AS (SELECT d.*, m.mad, CASE WHEN d.n >= 20 AND m.mad > 0 THEN d.dev / (1.4826 * m.mad) END AS zscore, (d.n >= 20 AND d.capacity_vph > 0 AND d.med_vol > 0 AND d.volume < 0.02 * d.med_vol) AS too_low FROM d JOIN m USING (link_class, area_type, period)) SELECT * FROM scored WHERE vc_ratio >= 0.95 OR (ABS(zscore) >= 3.0 AND COALESCE(vc_ratio, 0) >= 0.25) OR too_low ORDER BY vc_ratio DESC NULLS LAST, ABS(zscore) DESC NULLS LAST

[src: findings/8d04a74eff1c; link_flows.csv (link_flows) row 11236 col volume]

Likely cause: Demand concentrated on this link because parallel routes are missing or slow, or the capacity/lanes are under-coded.

[src: findings/8d04a74eff1c; link_flows.csv (link_flows) row 11236 col volume]

Suggested action: Check lanes and capacity coding on this link and its competitors; compare with counts if available.

[src: findings/8d04a74eff1c; link_flows.csv (link_flows) row 11236 col volume]

Evidence

Metric	Value	Threshold
issue	over_capacity	
volume	2,430	
vc_ratio	1.16	
zscore	1.415	
group_median_volume	850	
group_size	22	
link_class	COL	
area_type	CBD	
capacity_vph	700	
lanes	1	
too_low	no	
vc_critical		1.30
vc_high		1.10
vc_medium		0.95
zscore_threshold		3
min_group_size		20
low_volume_share_of_group_median		2.0%
min_vc_for_outlier		0.25

[src: findings/8d04a74eff1c; link_flows.csv (link_flows) row 11236 col volume]

Sources: link_flows.csv (link_flows) row 11236 col volume

[src: findings/8d04a74eff1c; link_flows.csv (link_flows) row 11236 col volume]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Add one lane	Add a lane to this link, raising capacity from 700 to 1,400 veh/h (one lane = 700 veh/h).	V/C falls from 1.16 to about 0.58 if demand is unchanged	sketch elasticity	medium
Verify capacity coding	Check that lanes and capacity per lane match the link's real cross-section; an under-coded capacity produces the same symptom as real congestion.	Not estimated	not computable	medium
Check for missing parallel routes	Confirm that parallel roads in the corridor are coded and open in this scenario; a closed or missing alternative concentrates demand here.	Not estimated	not computable	low

[src: findings/8d04a74eff1c; link_flows.csv (link_flows) row 11236 col volume]

Unrealistic speeds and times: Link 29 (LOC, 1016-1017), AM

High: Traffic on link 29 (LOC) crawls at 2.0 km/h in the AM peak.

Modeller view: Method: link_flows joined to links; thresholds on speed ratio, floor, delay

Values: issue=speed_below_floor, cong_speed_kph=2, ffs_kph=40, delay_s=0, volume=36, speed_ratio=0.05, link_class=LOC

Thresholds: min_cong_speed_kph=5, max_speed_over_ffs_ratio=1.05, max_delay_s=900

Files: link_flows.csv rows 2933

SQL: SELECT f.link_id, f.period, f.volume, f.cong_speed_kph, f.delay_s, f.cong_time_s, l.ffs_kph,

I.link_class, I.length_m, f.source_file, f.source_row, f.cong_speed_kph / NULLIF(I.ffc_kph, 0) AS speed_ratio FROM link_flows f JOIN links I USING (link_id) WHERE f.user_class = 'ALL' AND f.period IN ('AM', 'PM') AND ((f.cong_speed_kph IS NOT NULL AND I.ffc_kph > 0 AND f.cong_speed_kph > 1.05 * I.ffc_kph) OR (f.cong_speed_kph IS NOT NULL AND f.cong_speed_kph < 5.0) OR (f.delay_s IS NOT NULL AND f.delay_s > 900.0)) ORDER BY f.link_id, f.period

[src: findings/9030a67b3246; link_flows.csv (link_flows) row 2933 col cong_speed_kph]

Likely cause: Severe over-capacity or a volume-delay function that collapses at high V/C; sometimes a zero-length or zero-speed coding.

[src: findings/9030a67b3246; link_flows.csv (link_flows) row 2933 col cong_speed_kph]

Suggested action: Check V/C on this link, the volume-delay parameters for its class, and the free-flow speed coding.

[src: findings/9030a67b3246; link_flows.csv (link_flows) row 2933 col cong_speed_kph]

Evidence

Metric	Value	Threshold
issue	speed_below_floor	
issues	speed_below_floor	
cong_speed_kph	2 km/h	
ffc_kph	40	
delay_s	0	
volume	40	
speed_ratio	0.05	
link_class	LOC	
min_cong_speed_kph		5 km/h
max_speed_over_ffc_ratio		1.05
max_delay_s		900

[src: findings/9030a67b3246; link_flows.csv (link_flows) row 2933 col cong_speed_kph]

Sources: link_flows.csv (link_flows) row 2933 col cong_speed_kph

[src: findings/9030a67b3246; link_flows.csv (link_flows) row 2933 col cong_speed_kph]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Review volume-delay parameters	Check the BPR/conical parameters for this link class; extreme delay or near-zero speed usually means the curve explodes above V/C 1.2.	Not estimated	not computable	low
Resolve capacity on this link first	If a V/C finding exists for the same link, fixing it removes this symptom.	Not estimated	not computable	low

[src: findings/9030a67b3246; link_flows.csv (link_flows) row 2933 col cong_speed_kph]

Parameter drift: Parameter VOT_CAR

High: Parameter VOT_CAR is set to 45.0 but the approved value is 38.5.

Modeller view: Method: parameters.value vs approved_value (numeric or text)

Values: key=VOT_CAR, value=45.0, approved_value=38.5, change_share=0.1688, drifted=True

Thresholds: numeric_tolerance=1e-09

Files: parameters.csv rows 1

SQL: SELECT key, value, approved_value, source_file, source_row, TRY_CAST(value AS DOUBLE) AS v_num, TRY_CAST(approved_value AS DOUBLE) AS a_num FROM parameters WHERE approved_value IS NOT NULL AND CASE WHEN TRY_CAST(value AS DOUBLE) IS NOT NULL AND TRY_CAST(approved_value AS DOUBLE) IS NOT NULL THEN ABS(TRY_CAST(value AS DOUBLE) - TRY_CAST(approved_value AS DOUBLE)) > 1e-09 * GREATEST(1.0, ABS(TRY_CAST(approved_value

AS DOUBLE))) ELSE TRIM(COALESCE(value, "")) <> TRIM(approved_value) END ORDER BY key

[src: findings/cac15209b289; parameters.csv (parameters) row 1 col value]

Likely cause: Parameter edited for a test and not reverted, or the approved register is out of date.

[src: findings/cac15209b289; parameters.csv (parameters) row 1 col value]

Suggested action: Restore the approved value, or record the change and get the new value approved.

[src: findings/cac15209b289; parameters.csv (parameters) row 1 col value]

Evidence

Metric	Value	Threshold
key	VOT_CAR	
value	45.0	
approved_value	38.5	
change_share	16.9%	
drifted	yes	
numeric_tolerance		1e-09

[src: findings/cac15209b289; parameters.csv (parameters) row 1 col value]

Sources: parameters.csv (parameters) row 1 col value

[src: findings/cac15209b289; parameters.csv (parameters) row 1 col value]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Restore the approved value	Reset the parameter to its approved value or get the new value approved.	Not estimated	not computable	low

[src: findings/cac15209b289; parameters.csv (parameters) row 1 col value]

Centroid connector placement: Link 90002 (CONN, 16-1100)

High: Connector 90002 loads zone traffic straight onto a FWY link at node 1100.

Modeller view: Method: connector street-end node joined to links of high class

Values: issue=connector_on_high_class, street_node=1100, high_class_link=183, high_class=FWY, length_m=9483, centroid=16

Thresholds: high_class_links=['FWY', 'EXP']

Files: links.csv rows 969

SQL: WITH conn AS (SELECT l.link_id, l.a_node, l.b_node, l.length_m, l.link_class, l.source_file, l.source_row, CASE WHEN COALESCE(na.is_centroid, FALSE) THEN l.b_node ELSE l.a_node END AS street_node, CASE WHEN COALESCE(na.is_centroid, FALSE) THEN l.a_node WHEN COALESCE(nb.is_centroid, FALSE) THEN l.b_node END AS centroid FROM links l LEFT JOIN nodes na ON na.node_id = l.a_node LEFT JOIN nodes nb ON nb.node_id = l.b_node WHERE l.link_class IN ('CONN') OR COALESCE(na.is_centroid, FALSE) OR COALESCE(nb.is_centroid, FALSE)), touch AS (SELECT c.link_id, MIN(h.link_id) AS high_link, ANY_VALUE(h.link_class) AS high_class FROM conn c JOIN links h ON (h.a_node = c.street_node OR h.b_node = c.street_node) AND h.link_class IN ('FWY', 'EXP') GROUP BY c.link_id) SELECT c.*, t.high_link, t.high_class FROM conn c LEFT JOIN touch t USING (link_id) WHERE t.high_link IS NOT NULL OR c.length_m > 5000.0 ORDER BY c.link_id

[src: findings/d4bd8020d016; links.csv (links) row 969 col link_id]

Likely cause: Connector snapped to the nearest node, which happens to be on the freeway, rather than to a local street.

[src: findings/d4bd8020d016; links.csv (links) row 969 col link_id]

Suggested action: Move the connector to a local or collector street node near the zone centroid.

[src: findings/d4bd8020d016; links.csv (links) row 969 col link_id]

Evidence

Metric	Value	Threshold
issue	connector_on_high_class	
street_node	1,100	
high_class_link	183	
high_class	FWY	
length_m	9,483	
centroid	16	
high_class_links		FWY, EXP

[src: findings/d4bd8020d016; links.csv (links) row 969 col link_id]

Sources: links.csv (links) row 969 col link_id

[src: findings/d4bd8020d016; links.csv (links) row 969 col link_id]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Relocate the connector to a local street	Move connector 90002 from node 1100 to the nearest collector or local street node so zone traffic enters via ramps/junctions.	removes about 15 vehicles per day (1 veh/h in AM) from the FWY link 183	sketch elasticity	medium
Split the zone	If the zone straddles the freeway, split it so each part connects to its own side of the network.	Not estimated	not computable	low

[src: findings/d4bd8020d016; links.csv (links) row 969 col link_id]

Link volume outliers: Link 85 (ART, 1046-1047), AM

High: Link 85 (ART) carries only 12 vehicles in the AM peak, against a typical 1,136 for similar links.

Modeller view: Method: V/C from link_flows; robust $z = (\ln(v+1) - \text{median}) / (1.4826 \text{ MAD})$ within link_class x area_type x period

Values: issue=volume_too_low, volume=11.78, vc_ratio=0.0022, zscore=-7.779, group_median_volume=1136, group_size=80, link_class=ART, area_type=URB, capacity_vph=1800, lanes=2, too_low=True

Thresholds: vc_critical=1.3, vc_high=1.1, vc_medium=0.95, zscore_threshold=3, min_group_size=20, low_volume_share_of_group_median=0.02, min_vc_for_outlier=0.25

Files: link_flows.csv rows 2989

SQL: WITH ph(period, hours) AS (VALUES ('AM', 3.0), ('PM', 3.0)), f AS (SELECT f.link_id, f.period, f.volume, COALESCE(f.vc_ratio, f.volume / NULLIF(l.capacity_vph * ph.hours, 0)) AS vc_ratio, l.link_class, l.area_type, l.capacity_vph, l.lanes, f.source_file, f.source_row FROM link_flows f JOIN links l USING (link_id) JOIN ph ON ph.period = f.period WHERE f.user_class = 'ALL' AND f.volume IS NOT NULL AND l.link_class NOT IN ('CONN')), g AS (SELECT link_class, area_type, period, COUNT(*) AS n, MEDIAN(LN(volume + 1)) AS med_log, MEDIAN(volume) AS med_vol FROM f GROUP BY 1, 2, 3), d AS (SELECT f.*, g.n, g.med_log, g.med_vol, LN(f.volume + 1) - g.med_log AS dev FROM f JOIN g USING (link_class, area_type, period)), m AS (SELECT link_class, area_type, period, MEDIAN(ABS(dev)) AS mad FROM d GROUP BY 1, 2, 3), scored AS (SELECT d.*, m.mad, CASE WHEN d.n >= 20 AND m.mad > 0 THEN d.dev / (1.4826 * m.mad) END AS zscore, (d.n >= 20 AND d.capacity_vph > 0 AND d.med_vol > 0 AND d.volume < 0.02 * d.med_vol) AS too_low FROM d JOIN m USING (link_class, area_type, period)) SELECT * FROM scored WHERE vc_ratio >= 0.95 OR (ABS(zscore) >= 3.0 AND COALESCE(vc_ratio, 0) >= 0.25) OR too_low ORDER BY vc_ratio DESC NULLS LAST, ABS(zscore) DESC NULLS LAST

[src: findings/ed9e8052b5b7; link_flows.csv (link_flows) row 2989 col volume]

Likely cause: Link is effectively unreachable (one-way coding, missing turn, or a disconnected end), or a parallel link carries its traffic.

[src: findings/ed9e8052b5b7; link_flows.csv (link_flows) row 2989 col volume]

Suggested action: Trace the path to this link and check one-way and turn coding.

[src: findings/ed9e8052b5b7; link_flows.csv (link_flows) row 2989 col volume]

Evidence

Metric	Value	Threshold
issue	volume_too_low	
volume	10	
vc_ratio	0.00	
zscore	-7.779	
group_median_volume	1,140	
group_size	80	
link_class	ART	
area_type	URB	
capacity_vph	1,800	
lanes	2	
too_low	yes	
vc_critical		1.30
vc_high		1.10
vc_medium		0.95
zscore_threshold		3
min_group_size		20
low_volume_share_of_group_median		2.0%
min_vc_for_outlier		0.25

[src: findings/ed9e8052b5b7; link_flows.csv (link_flows) row 2989 col volume]

Sources: link_flows.csv (link_flows) row 2989 col volume

[src: findings/ed9e8052b5b7; link_flows.csv (link_flows) row 2989 col volume]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Trace access to the link	Build a path to this link in the assignment network; a one-way flag, missing turn or disconnected end usually explains near-zero volume.	Not estimated	not computable	low
Check for missing parallel routes	Confirm that parallel roads in the corridor are coded and open in this scenario; a closed or missing alternative concentrates demand here.	Not estimated	not computable	low

[src: findings/ed9e8052b5b7; link_flows.csv (link_flows) row 2989 col volume]

Link volume outliers: Link 587 (COL, 1107-1121), AM

High: Link 587 (COL) carries 2,384 vehicles in the AM peak, 1.14 times its capacity.

Modeller view: Method: V/C from link_flows; robust $z = (\ln(v+1) - \text{median}) / (1.4826 \text{ MAD})$ within link_class x area_type x period

Values: issue=over_capacity, volume=2384, vc_ratio=1.135, zscore=1.386, group_median_volume=690.9, group_size=22, link_class=COL, area_type=CBD, capacity_vph=700, lanes=1, too_low=False

Thresholds: vc_critical=1.3, vc_high=1.1, vc_medium=0.95, zscore_threshold=3, min_group_size=20, low_volume_share_of_group_median=0.02, min_vc_for_outlier=0.25

Files: link_flows.csv rows 3491

SQL: WITH ph(period, hours) AS (VALUES ('AM', 3.0), ('PM', 3.0)), f AS (SELECT f.link_id, f.period, f.volume, COALESCE(f.vc_ratio, f.volume / NULLIF(l.capacity_vph * ph.hours, 0)) AS vc_ratio, l.link_class,

```

l.area_type, l.capacity_vph, l.lanes, f.source_file, f.source_row FROM link_flows f JOIN links l USING
(link_id) JOIN ph ON ph.period = f.period WHERE f.user_class = 'ALL' AND f.volume IS NOT NULL AND
l.link_class NOT IN ('CONN') ), g AS ( SELECT link_class, area_type, period, COUNT(*) AS n,
MEDIAN(LN(volume + 1)) AS med_log, MEDIAN(volume) AS med_vol FROM f GROUP BY 1, 2, 3 ), d AS
( SELECT f.*, g.n, g.med_log, g.med_vol, LN(f.volume + 1) - g.med_log AS dev FROM f JOIN g USING
(link_class, area_type, period) ), m AS ( SELECT link_class, area_type, period, MEDIAN(ABS(dev)) AS
mad FROM d GROUP BY 1, 2, 3 ), scored AS ( SELECT d.*, m.mad, CASE WHEN d.n >= 20 AND m.mad
> 0 THEN d.dev / (1.4826 * m.mad) END AS zscore, (d.n >= 20 AND d.capacity_vph > 0 AND d.med_vol >
0 AND d.volume < 0.02 * d.med_vol) AS too_low FROM d JOIN m USING (link_class, area_type, period) )
SELECT * FROM scored WHERE vc_ratio >= 0.95 OR (ABS(zscore) >= 3.0 AND COALESCE(vc_ratio, 0)
>= 0.25) OR too_low ORDER BY vc_ratio DESC NULLS LAST, ABS(zscore) DESC NULLS LAST

```

[src: findings/f5416817f042; link_flows.csv (link_flows) row 3491 col volume]

Likely cause: Demand concentrated on this link because parallel routes are missing or slow, or the capacity/lanes are under-coded.

[src: findings/f5416817f042; link_flows.csv (link_flows) row 3491 col volume]

Suggested action: Check lanes and capacity coding on this link and its competitors; compare with counts if available.

[src: findings/f5416817f042; link_flows.csv (link_flows) row 3491 col volume]

Evidence

Metric	Value	Threshold
issue	over_capacity	
volume	2,380	
vc_ratio	1.14	
zscore	1.386	
group_median_volume	690	
group_size	22	
link_class	COL	
area_type	CBD	
capacity_vph	700	
lanes	1	
too_low	no	
vc_critical		1.30
vc_high		1.10
vc_medium		0.95
zscore_threshold		3
min_group_size		20
low_volume_share_of_group_median		2.0%
min_vc_for_outlier		0.25

[src: findings/f5416817f042; link_flows.csv (link_flows) row 3491 col volume]

Sources: link_flows.csv (link_flows) row 3491 col volume

[src: findings/f5416817f042; link_flows.csv (link_flows) row 3491 col volume]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Add one lane	Add a lane to this link, raising capacity from 700 to 1,400 veh/h (one lane = 700 veh/h).	V/C falls from 1.14 to about 0.57 if demand is unchanged	sketch elasticity	medium

Measure	Description	Estimated effect	Method	Confidence
Verify capacity coding	Check that lanes and capacity per lane match the link's real cross-section; an under-coded capacity produces the same symptom as real congestion.	Not estimated	not computable	medium
Check for missing parallel routes	Confirm that parallel roads in the corridor are coded and open in this scenario; a closed or missing alternative concentrates demand here.	Not estimated	not computable	low

[src: findings/f5416817f042; link_flows.csv (link_flows) row 3491 col volume]

Transit line integrity: Line BUS_2 (BUS), AM

High: Line BUS_2 (BUS) runs every 0 minutes in the AM peak, below the 2 minute minimum.

Modeller view: Method: headway range and per-mode maximum

Values: issue=headway_out_of_range, headway_min=0.5, mode=BUS, vehicle_capacity=80

Thresholds: headway_min_range=[2.0, 120.0], mode_max_headway_min=120

Files: transit_lines.csv rows 16

SQL: SELECT line_id, mode, period, headway_min, vehicle_capacity, source_file, source_row FROM transit_lines WHERE headway_min IS NOT NULL ORDER BY line_id, period

[src: findings/fd51f22f77b5; transit_lines.csv (transit_lines) row 16 col headway_min]

Likely cause: Headway coded in the wrong unit (seconds or vehicles per hour) or a placeholder value left in the line file.

[src: findings/fd51f22f77b5; transit_lines.csv (transit_lines) row 16 col headway_min]

Suggested action: Correct the headway in the line file; a typical BUS service runs every 2-120 minutes.

[src: findings/fd51f22f77b5; transit_lines.csv (transit_lines) row 16 col headway_min]

Evidence

Metric	Value	Threshold
issue	headway_out_of_range	
headway_min	0.5	
mode	BUS	
vehicle_capacity	80	
headway_min_range		2, 120
mode_max_headway_min		120

[src: findings/fd51f22f77b5; transit_lines.csv (transit_lines) row 16 col headway_min]

Sources: transit_lines.csv (transit_lines) row 16 col headway_min

[src: findings/fd51f22f77b5; transit_lines.csv (transit_lines) row 16 col headway_min]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Set headway to 2 minutes	Replace the 0.5 minute headway with the BUS limit of 2 minutes (or the intended service frequency).	seat-km per hour changes from 364,552 to 91,138	sketch elasticity	medium
Check the headway unit	A value like this often means seconds or vehicles per hour were exported instead of minutes.	Not estimated	not computable	low

[src: findings/fd51f22f77b5; transit_lines.csv (transit_lines) row 16 col headway_min]

Transit line integrity: Line BUS_2 (BUS), NT

High: Line BUS_2 (BUS) runs every 240 minutes in the night period, above the 120 minute maximum for BUS.

Modeller view: Method: headway range and per-mode maximum

Values: issue=headway_out_of_range, headway_min=240, mode=BUS, vehicle_capacity=80

Thresholds: headway_min_range=[2.0, 120.0], mode_max_headway_min=120

Files: transit_lines.csv rows 20

SQL: SELECT line_id, mode, period, headway_min, vehicle_capacity, source_file, source_row FROM transit_lines WHERE headway_min IS NOT NULL ORDER BY line_id, period

[src: findings/ff39ab292787; transit_lines.csv (transit_lines) row 20 col headway_min]

Likely cause: Headway coded in the wrong unit (seconds or vehicles per hour) or a placeholder value left in the line file.

[src: findings/ff39ab292787; transit_lines.csv (transit_lines) row 20 col headway_min]

Suggested action: Correct the headway in the line file; a typical BUS service runs every 2-120 minutes.

[src: findings/ff39ab292787; transit_lines.csv (transit_lines) row 20 col headway_min]

Evidence

Metric	Value	Threshold
issue	headway_out_of_range	
headway_min	240	
mode	BUS	
vehicle_capacity	80	
headway_min_range		2, 120
mode_max_headway_min		120

[src: findings/ff39ab292787; transit_lines.csv (transit_lines) row 20 col headway_min]

Sources: transit_lines.csv (transit_lines) row 20 col headway_min

[src: findings/ff39ab292787; transit_lines.csv (transit_lines) row 20 col headway_min]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Set headway to 120 minutes	Replace the 240.0 minute headway with the BUS limit of 120 minutes (or the intended service frequency).	seat-km per hour changes from 759 to 1,519	sketch elasticity	medium
Check the headway unit	A value like this often means seconds or vehicles per hour were exported instead of minutes.	Not estimated	not computable	low

[src: findings/ff39ab292787; transit_lines.csv (transit_lines) row 20 col headway_min]

Medium findings (13)

Link volume outliers: Link 190 (FWY, 1103-1102), PM

Medium: Link 190 (FWY, URB) carries 7,865 vehicles in the PM peak, far above similar links (typical 5,383).

Modeller view: Method: V/C from link_flows; robust $z = (\ln(v+1) - \text{median}) / (1.4826 \text{ MAD})$ within link_class x area_type x period

Values: issue=statistical_outlier, volume=7865, vc_ratio=0.4369, zscore=3.37,

group_median_volume=5383, group_size=40, link_class=FWY, area_type=URB, capacity_vph=6000, lanes=3, too_low=False

Thresholds: vc_critical=1.3, vc_high=1.1, vc_medium=0.95, zscore_threshold=3, min_group_size=20, low_volume_share_of_group_median=0.02, min_vc_for_outlier=0.25

Files: link_flows.csv rows 10838

SQL: WITH ph(period, hours) AS (VALUES ('AM', 3.0), ('PM', 3.0)), f AS (SELECT f.link_id, f.period, f.volume, COALESCE(f.vc_ratio, f.volume / NULLIF(l.capacity_vph * ph.hours, 0)) AS vc_ratio, l.link_class, l.area_type, l.capacity_vph, l.lanes, f.source_file, f.source_row FROM link_flows f JOIN links l USING

```
(link_id) JOIN ph ON ph.period = f.period WHERE f.user_class = 'ALL' AND f.volume IS NOT NULL AND
l.link_class NOT IN ('CONN') ), g AS ( SELECT link_class, area_type, period, COUNT(*) AS n,
MEDIAN(LN(volume + 1)) AS med_log, MEDIAN(volume) AS med_vol FROM f GROUP BY 1, 2, 3 ), d AS
( SELECT f.*, g.n, g.med_log, g.med_vol, LN(f.volume + 1) - g.med_log AS dev FROM f JOIN g USING
(link_class, area_type, period) ), m AS ( SELECT link_class, area_type, period, MEDIAN(ABS(dev)) AS
mad FROM d GROUP BY 1, 2, 3 ), scored AS ( SELECT d.*, m.mad, CASE WHEN d.n >= 20 AND m.mad
> 0 THEN d.dev / (1.4826 * m.mad) END AS zscore, (d.n >= 20 AND d.capacity_vph > 0 AND d.med_vol >
0 AND d.volume < 0.02 * d.med_vol) AS too_low FROM d JOIN m USING (link_class, area_type, period) )
SELECT * FROM scored WHERE vc_ratio >= 0.95 OR (ABS(zscore) >= 3.0 AND COALESCE(vc_ratio, 0)
>= 0.25) OR too_low ORDER BY vc_ratio DESC NULLS LAST, ABS(zscore) DESC NULLS LAST
```

[src: findings/16b892734726; link_flows.csv (link_flows) row 10838 col volume]

Likely cause: Attribute mis-coding (class, speed or capacity) making the link unusually attractive or unattractive.

[src: findings/16b892734726; link_flows.csv (link_flows) row 10838 col volume]

Suggested action: Review link attributes against neighbouring links of the same class.

[src: findings/16b892734726; link_flows.csv (link_flows) row 10838 col volume]

Evidence

Metric	Value	Threshold
issue	statistical_outlier	
volume	7,860	
vc_ratio	0.44	
zscore	3.37	
group_median_volume	5,380	
group_size	40	
link_class	FWY	
area_type	URB	
capacity_vph	6,000	
lanes	3	
too_low	no	
vc_critical		1.30
vc_high		1.10
vc_medium		0.95
zscore_threshold		3
min_group_size		20
low_volume_share_of_group_median		2.0%
min_vc_for_outlier		0.25

[src: findings/16b892734726; link_flows.csv (link_flows) row 10838 col volume]

Sources: link_flows.csv (link_flows) row 10838 col volume

[src: findings/16b892734726; link_flows.csv (link_flows) row 10838 col volume]

Link volume outliers: Link 535 (ART, 1105-1119), AM

Medium: Link 535 (ART) carries 5,796 vehicles in the AM peak, 1.07 times its capacity.

Modeller view: Method: V/C from link_flows; robust $z = (\ln(v+1) - \text{median}) / (1.4826 \text{ MAD})$ within link_class x area_type x period

Values: issue=over_capacity, volume=5796, vc_ratio=1.073, zscore=1.262, group_median_volume=1645, group_size=30, link_class=ART, area_type=CBD, capacity_vph=1800, lanes=2, too_low=False

Thresholds: vc_critical=1.3, vc_high=1.1, vc_medium=0.95, zscore_threshold=3, min_group_size=20,

low_volume_share_of_group_median=0.02, min_vc_for_outlier=0.25

Files: link_flows.csv rows 3439

SQL: WITH ph(period, hours) AS (VALUES ('AM', 3.0), ('PM', 3.0)), f AS (SELECT f.link_id, f.period, f.volume, COALESCE(f.vc_ratio, f.volume / NULLIF(l.capacity_vph * ph.hours, 0)) AS vc_ratio, l.link_class, l.area_type, l.capacity_vph, l.lanes, f.source_file, f.source_row FROM link_flows f JOIN links l USING (link_id) JOIN ph ON ph.period = f.period WHERE f.user_class = 'ALL' AND f.volume IS NOT NULL AND l.link_class NOT IN ('CONN')), g AS (SELECT link_class, area_type, period, COUNT(*) AS n, MEDIAN(LN(volume + 1)) AS med_log, MEDIAN(volume) AS med_vol FROM f GROUP BY 1, 2, 3), d AS (SELECT f.*, g.n, g.med_log, g.med_vol, LN(f.volume + 1) - g.med_log AS dev FROM f JOIN g USING (link_class, area_type, period)), m AS (SELECT link_class, area_type, period, MEDIAN(ABS(dev)) AS mad FROM d GROUP BY 1, 2, 3), scored AS (SELECT d.*, m.mad, CASE WHEN d.n >= 20 AND m.mad > 0 THEN d.dev / (1.4826 * m.mad) END AS zscore, (d.n >= 20 AND d.capacity_vph > 0 AND d.med_vol > 0 AND d.volume < 0.02 * d.med_vol) AS too_low FROM d JOIN m USING (link_class, area_type, period)) SELECT * FROM scored WHERE vc_ratio >= 0.95 OR (ABS(zscore) >= 3.0 AND COALESCE(vc_ratio, 0) >= 0.25) OR too_low ORDER BY vc_ratio DESC NULLS LAST, ABS(zscore) DESC NULLS LAST

[src: findings/8a1351e9d62a; link_flows.csv (link_flows) row 3439 col volume]

Likely cause: Demand concentrated on this link because parallel routes are missing or slow, or the capacity/lanes are under-coded.

[src: findings/8a1351e9d62a; link_flows.csv (link_flows) row 3439 col volume]

Suggested action: Check lanes and capacity coding on this link and its competitors; compare with counts if available.

[src: findings/8a1351e9d62a; link_flows.csv (link_flows) row 3439 col volume]

Evidence

Metric	Value	Threshold
issue	over_capacity	
volume	5,800	
vc_ratio	1.07	
zscore	1.262	
group_median_volume	1,650	
group_size	30	
link_class	ART	
area_type	CBD	
capacity_vph	1,800	
lanes	2	
too_low	no	
vc_critical		1.30
vc_high		1.10
vc_medium		0.95
zscore_threshold		3
min_group_size		20
low_volume_share_of_group_median		2.0%
min_vc_for_outlier		0.25

[src: findings/8a1351e9d62a; link_flows.csv (link_flows) row 3439 col volume]

Sources: link_flows.csv (link_flows) row 3439 col volume

[src: findings/8a1351e9d62a; link_flows.csv (link_flows) row 3439 col volume]

Centroid connector placement: Link 90002 (CONN, 16-1100)

Medium: Connector 90002 is 9.5 km long, above the 5.0 km limit.

Modeller view: Method: connector length_m above threshold

Values: issue=connector_too_long, length_m=9483, centroid=16

Thresholds: max_connector_length_m=5000

Files: links.csv rows 969

SQL: WITH conn AS (SELECT l.link_id, l.a_node, l.b_node, l.length_m, l.link_class, l.source_file, l.source_row, CASE WHEN COALESCE(na.is_centroid, FALSE) THEN l.b_node ELSE l.a_node END AS street_node, CASE WHEN COALESCE(na.is_centroid, FALSE) THEN l.a_node WHEN COALESCE(nb.is_centroid, FALSE) THEN l.b_node END AS centroid FROM links l LEFT JOIN nodes na ON na.node_id = l.a_node LEFT JOIN nodes nb ON nb.node_id = l.b_node WHERE l.link_class IN ('CONN') OR COALESCE(na.is_centroid, FALSE) OR COALESCE(nb.is_centroid, FALSE)), touch AS (SELECT c.link_id, MIN(h.link_id) AS high_link, ANY_VALUE(h.link_class) AS high_class FROM conn c JOIN links h ON (h.a_node = c.street_node OR h.b_node = c.street_node) AND h.link_class IN ('FWY', 'EXP') GROUP BY c.link_id) SELECT c.*, t.high_link, t.high_class FROM conn c LEFT JOIN touch t USING (link_id) WHERE t.high_link IS NOT NULL OR c.length_m > 5000.0 ORDER BY c.link_id

[src: findings/8c9efe2ce4bd; links.csv (links) row 969 col link_id]

Likely cause: Zone centroid far from the loaded network, or the connector length was left at a default.

[src: findings/8c9efe2ce4bd; links.csv (links) row 969 col link_id]

Suggested action: Split the zone, add local streets, or connect to a closer node.

[src: findings/8c9efe2ce4bd; links.csv (links) row 969 col link_id]

Evidence

Metric	Value	Threshold
issue	connector_too_long	
length_m	9,483	
centroid	16	
max_connector_length_m		5,000

[src: findings/8c9efe2ce4bd; links.csv (links) row 969 col link_id]

Sources: links.csv (links) row 969 col link_id

[src: findings/8c9efe2ce4bd; links.csv (links) row 969 col link_id]

Network connectivity: Node 9003

Medium: Node 9003 is not connected to any road link.

Modeller view: Method: nodes anti-joined to link end nodes

Values: issue=orphan_node, node_id=9003

Files: nodes.csv rows 259

SQL: SELECT n.node_id, n.x, n.y, n.source_file, n.source_row FROM nodes n LEFT JOIN (SELECT a_node AS node_id FROM links UNION SELECT b_node FROM links) u USING (node_id) WHERE u.node_id IS NULL AND NOT COALESCE(n.is_centroid, FALSE) ORDER BY n.node_id

[src: findings/8e6221b48abb; nodes.csv (nodes) row 259 col node_id]

Likely cause: Leftover node from network editing, or links deleted around it.

[src: findings/8e6221b48abb; nodes.csv (nodes) row 259 col node_id]

Suggested action: Delete the node or connect it; orphan nodes are harmless for assignment but signal an incomplete edit.

[src: findings/8e6221b48abb; nodes.csv (nodes) row 259 col node_id]

Evidence

Metric	Value	Threshold
issue	orphan_node	
node_id	9,003	

[src: findings/8e6221b48abb; nodes.csv (nodes) row 259 col node_id]

Sources: nodes.csv (nodes) row 259 col node_id

[src: findings/8e6221b48abb; nodes.csv (nodes) row 259 col node_id]

Link volume outliers: Link 536 (ART, 1119-1105), PM

Medium: Link 536 (ART) carries 5,744 vehicles in the PM peak, 1.06 times its capacity.

Modeller view: Method: V/C from link_flows; robust $z = (\ln(v+1) - \text{median}) / (1.4826 \text{ MAD})$ within link_class x area_type x period

Values: issue=over_capacity, volume=5744, vc_ratio=1.064, zscore=1.135, group_median_volume=1766, group_size=30, link_class=ART, area_type=CBD, capacity_vph=1800, lanes=2, too_low=False

Thresholds: vc_critical=1.3, vc_high=1.1, vc_medium=0.95, zscore_threshold=3, min_group_size=20, low_volume_share_of_group_median=0.02, min_vc_for_outlier=0.25

Files: link_flows.csv rows 11184

SQL: WITH ph(period, hours) AS (VALUES ('AM', 3.0), ('PM', 3.0)), f AS (SELECT f.link_id, f.period, f.volume, COALESCE(f.vc_ratio, f.volume / NULLIF(l.capacity_vph * ph.hours, 0)) AS vc_ratio, l.link_class, l.area_type, l.capacity_vph, l.lanes, f.source_file, f.source_row FROM link_flows f JOIN links l USING (link_id) JOIN ph ON ph.period = f.period WHERE f.user_class = 'ALL' AND f.volume IS NOT NULL AND l.link_class NOT IN ('CONN')), g AS (SELECT link_class, area_type, period, COUNT(*) AS n, MEDIAN(LN(volume + 1)) AS med_log, MEDIAN(volume) AS med_vol FROM f GROUP BY 1, 2, 3), d AS (SELECT f.*, g.n, g.med_log, g.med_vol, LN(f.volume + 1) - g.med_log AS dev FROM f JOIN g USING (link_class, area_type, period)), m AS (SELECT link_class, area_type, period, MEDIAN(ABS(dev)) AS mad FROM d GROUP BY 1, 2, 3), scored AS (SELECT d.*, m.mad, CASE WHEN d.n >= 20 AND m.mad > 0 THEN d.dev / (1.4826 * m.mad) END AS zscore, (d.n >= 20 AND d.capacity_vph > 0 AND d.med_vol > 0 AND d.volume < 0.02 * d.med_vol) AS too_low FROM d JOIN m USING (link_class, area_type, period)) SELECT * FROM scored WHERE vc_ratio >= 0.95 OR (ABS(zscore) >= 3.0 AND COALESCE(vc_ratio, 0) >= 0.25) OR too_low ORDER BY vc_ratio DESC NULLS LAST, ABS(zscore) DESC NULLS LAST

[src: findings/957dfa057e28; link_flows.csv (link_flows) row 11184 col volume]

Likely cause: Demand concentrated on this link because parallel routes are missing or slow, or the capacity/lanes are under-coded.

[src: findings/957dfa057e28; link_flows.csv (link_flows) row 11184 col volume]

Suggested action: Check lanes and capacity coding on this link and its competitors; compare with counts if available.

[src: findings/957dfa057e28; link_flows.csv (link_flows) row 11184 col volume]

Evidence

Metric	Value	Threshold
issue	over_capacity	
volume	5,740	
vc_ratio	1.06	
zscore	1.135	
group_median_volume	1,770	
group_size	30	
link_class	ART	
area_type	CBD	
capacity_vph	1,800	
lanes	2	
too_low	no	
vc_critical		1.30
vc_high		1.10
vc_medium		0.95

Metric	Value	Threshold
zscore_threshold		3
min_group_size		20
low_volume_share_of_group_median		2.0%
min_vc_for_outlier		0.25

[src: findings/957dfa057e28; link_flows.csv (link_flows) row 11184 col volume]

Sources: link_flows.csv (link_flows) row 11184 col volume

[src: findings/957dfa057e28; link_flows.csv (link_flows) row 11184 col volume]

Network connectivity: Node 9001

Medium: Node 9001 is not connected to any road link.

Modeller view: Method: nodes anti-joined to link end nodes

Values: issue=orphan_node, node_id=9001

Files: nodes.csv rows 257

SQL: SELECT n.node_id, n.x, n.y, n.source_file, n.source_row FROM nodes n LEFT JOIN (SELECT a_node AS node_id FROM links UNION SELECT b_node FROM links) u USING (node_id) WHERE u.node_id IS NULL AND NOT COALESCE(n.is_centroid, FALSE) ORDER BY n.node_id

[src: findings/a4aeaf5e2cf9; nodes.csv (nodes) row 257 col node_id]

Likely cause: Leftover node from network editing, or links deleted around it.

[src: findings/a4aeaf5e2cf9; nodes.csv (nodes) row 257 col node_id]

Suggested action: Delete the node or connect it; orphan nodes are harmless for assignment but signal an incomplete edit.

[src: findings/a4aeaf5e2cf9; nodes.csv (nodes) row 257 col node_id]

Evidence

Metric	Value	Threshold
issue	orphan_node	
node_id	9,001	

[src: findings/a4aeaf5e2cf9; nodes.csv (nodes) row 257 col node_id]

Sources: nodes.csv (nodes) row 257 col node_id

[src: findings/a4aeaf5e2cf9; nodes.csv (nodes) row 257 col node_id]

Link volume outliers: Link 587 (COL, 1107-1121), PM

Medium: Link 587 (COL) carries 2,122 vehicles in the PM peak, 1.01 times its capacity.

Modeller view: Method: V/C from link_flows; robust $z = (\ln(v+1) - \text{median}) / (1.4826 \text{ MAD})$ within link_class x area_type x period

Values: issue=over_capacity, volume=2122, vc_ratio=1.01, zscore=1.237, group_median_volume=851.4, group_size=22, link_class=COL, area_type=CBD, capacity_vph=700, lanes=1, too_low=False

Thresholds: vc_critical=1.3, vc_high=1.1, vc_medium=0.95, zscore_threshold=3, min_group_size=20, low_volume_share_of_group_median=0.02, min_vc_for_outlier=0.25

Files: link_flows.csv rows 11235

SQL: WITH ph(period, hours) AS (VALUES ('AM', 3.0), ('PM', 3.0)), f AS (SELECT f.link_id, f.period, f.volume, COALESCE(f.vc_ratio, f.volume / NULLIF(l.capacity_vph * ph.hours, 0)) AS vc_ratio, l.link_class, l.area_type, l.capacity_vph, l.lanes, f.source_file, f.source_row FROM link_flows f JOIN links l USING (link_id) JOIN ph ON ph.period = f.period WHERE f.user_class = 'ALL' AND f.volume IS NOT NULL AND l.link_class NOT IN ('CONN')), g AS (SELECT link_class, area_type, period, COUNT(*) AS n, MEDIAN(LN(volume + 1)) AS med_log, MEDIAN(volume) AS med_vol FROM f GROUP BY 1, 2, 3), d AS (SELECT f.*, g.n, g.med_log, g.med_vol, LN(f.volume + 1) - g.med_log AS dev FROM f JOIN g USING (link_class, area_type, period)), m AS (SELECT link_class, area_type, period, MEDIAN(ABS(dev)) AS

mad FROM d GROUP BY 1, 2, 3), scored AS (SELECT d.*, m.mad, CASE WHEN d.n >= 20 AND m.mad > 0 THEN d.dev / (1.4826 * m.mad) END AS zscore, (d.n >= 20 AND d.capacity_vph > 0 AND d.med_vol > 0 AND d.volume < 0.02 * d.med_vol) AS too_low FROM d JOIN m USING (link_class, area_type, period)) SELECT * FROM scored WHERE vc_ratio >= 0.95 OR (ABS(zscore) >= 3.0 AND COALESCE(vc_ratio, 0) >= 0.25) OR too_low ORDER BY vc_ratio DESC NULLS LAST, ABS(zscore) DESC NULLS LAST

[src: findings/a68d84d9aefe; link_flows.csv (link_flows) row 11235 col volume]

Likely cause: Demand concentrated on this link because parallel routes are missing or slow, or the capacity/lanes are under-coded.

[src: findings/a68d84d9aefe; link_flows.csv (link_flows) row 11235 col volume]

Suggested action: Check lanes and capacity coding on this link and its competitors; compare with counts if available.

[src: findings/a68d84d9aefe; link_flows.csv (link_flows) row 11235 col volume]

Evidence

Metric	Value	Threshold
issue	over_capacity	
volume	2,120	
vc_ratio	1.01	
zscore	1.237	
group_median_volume	850	
group_size	22	
link_class	COL	
area_type	CBD	
capacity_vph	700	
lanes	1	
too_low	no	
vc_critical		1.30
vc_high		1.10
vc_medium		0.95
zscore_threshold		3
min_group_size		20
low_volume_share_of_group_median		2.0%
min_vc_for_outlier		0.25

[src: findings/a68d84d9aefe; link_flows.csv (link_flows) row 11235 col volume]

Sources: link_flows.csv (link_flows) row 11235 col volume

[src: findings/a68d84d9aefe; link_flows.csv (link_flows) row 11235 col volume]

Network connectivity: Node 9002

Medium: Node 9002 is not connected to any road link.

Modeller view: Method: nodes anti-joined to link end nodes

Values: issue=orphan_node, node_id=9002

Files: nodes.csv rows 258

SQL: SELECT n.node_id, n.x, n.y, n.source_file, n.source_row FROM nodes n LEFT JOIN (SELECT a_node AS node_id FROM links UNION SELECT b_node FROM links) u USING (node_id) WHERE u.node_id IS NULL AND NOT COALESCE(n.is_centroid, FALSE) ORDER BY n.node_id

[src: findings/c0928a4b7772; nodes.csv (nodes) row 258 col node_id]

Likely cause: Leftover node from network editing, or links deleted around it.

[src: findings/c0928a4b7772; nodes.csv (nodes) row 258 col node_id]

Suggested action: Delete the node or connect it; orphan nodes are harmless for assignment but signal an incomplete edit.

[src: findings/c0928a4b7772; nodes.csv (nodes) row 258 col node_id]

Evidence

Metric	Value	Threshold
issue	orphan_node	
node_id	9,002	

[src: findings/c0928a4b7772; nodes.csv (nodes) row 258 col node_id]

Sources: nodes.csv (nodes) row 258 col node_id

[src: findings/c0928a4b7772; nodes.csv (nodes) row 258 col node_id]

Link volume outliers: Link 692 (ART, 1125-1111), AM

Medium: Link 692 (ART, SUB) carries 1,985 vehicles in the AM peak, far above similar links (typical 619).

Modeller view: Method: V/C from link_flows; robust $z = (\ln(v+1) - \text{median}) / (1.4826 \text{ MAD})$ within link_class x area_type x period

Values: issue=statistical_outlier, volume=1985, vc_ratio=0.3675, zscore=3.102, group_median_volume=618.6, group_size=92, link_class=ART, area_type=SUB, capacity_vph=1800, lanes=2, too_low=False

Thresholds: vc_critical=1.3, vc_high=1.1, vc_medium=0.95, zscore_threshold=3, min_group_size=20, low_volume_share_of_group_median=0.02, min_vc_for_outlier=0.25

Files: link_flows.csv rows 3596

SQL: WITH ph(period, hours) AS (VALUES ('AM', 3.0), ('PM', 3.0)), f AS (SELECT f.link_id, f.period, f.volume, COALESCE(f.vc_ratio, f.volume / NULLIF(l.capacity_vph * ph.hours, 0)) AS vc_ratio, l.link_class, l.area_type, l.capacity_vph, l.lanes, f.source_file, f.source_row FROM link_flows f JOIN links l USING (link_id) JOIN ph ON ph.period = f.period WHERE f.user_class = 'ALL' AND f.volume IS NOT NULL AND l.link_class NOT IN ('CONN')), g AS (SELECT link_class, area_type, period, COUNT(*) AS n, MEDIAN(LN(volume + 1)) AS med_log, MEDIAN(volume) AS med_vol FROM f GROUP BY 1, 2, 3), d AS (SELECT f.*, g.n, g.med_log, g.med_vol, LN(f.volume + 1) - g.med_log AS dev FROM f JOIN g USING (link_class, area_type, period)), m AS (SELECT link_class, area_type, period, MEDIAN(ABS(dev)) AS mad FROM d GROUP BY 1, 2, 3), scored AS (SELECT d.*, m.mad, CASE WHEN d.n >= 20 AND m.mad > 0 THEN d.dev / (1.4826 * m.mad) END AS zscore, (d.n >= 20 AND d.capacity_vph > 0 AND d.med_vol > 0 AND d.volume < 0.02 * d.med_vol) AS too_low FROM d JOIN m USING (link_class, area_type, period)) SELECT * FROM scored WHERE vc_ratio >= 0.95 OR (ABS(zscore) >= 3.0 AND COALESCE(vc_ratio, 0) >= 0.25) OR too_low ORDER BY vc_ratio DESC NULLS LAST, ABS(zscore) DESC NULLS LAST

[src: findings/c638a1090952; link_flows.csv (link_flows) row 3596 col volume]

Likely cause: Attribute mis-coding (class, speed or capacity) making the link unusually attractive or unattractive.

[src: findings/c638a1090952; link_flows.csv (link_flows) row 3596 col volume]

Suggested action: Review link attributes against neighbouring links of the same class.

[src: findings/c638a1090952; link_flows.csv (link_flows) row 3596 col volume]

Evidence

Metric	Value	Threshold
issue	statistical_outlier	
volume	1,980	
vc_ratio	0.37	
zscore	3.102	
group_median_volume	620	
group_size	92	
link_class	ART	

Metric	Value	Threshold
area_type	SUB	
capacity_vph	1,800	
lanes	2	
too_low	no	
vc_critical		1.30
vc_high		1.10
vc_medium		0.95
zscore_threshold		3
min_group_size		20
low_volume_share_of_group_median		2.0%
min_vc_for_outlier		0.25

[src: findings/c638a1090952; link_flows.csv (link_flows) row 3596 col volume]

Sources: link_flows.csv (link_flows) row 3596 col volume

[src: findings/c638a1090952; link_flows.csv (link_flows) row 3596 col volume]

Link volume outliers: Link 705 (LOC, 1028-1042), AM

Medium: Link 705 (LOC, RUR) carries 882 vehicles in the AM peak, far above similar links (typical 65).

Modeller view: Method: V/C from link_flows; robust $z = (\ln(v+1) - \text{median}) / (1.4826 \text{ MAD})$ within link_class x area_type x period

Values: issue=statistical_outlier, volume=882, vc_ratio=0.588, zscore=3.012, group_median_volume=65.14, group_size=48, link_class=LOC, area_type=RUR, capacity_vph=500, lanes=1, too_low=False

Thresholds: vc_critical=1.3, vc_high=1.1, vc_medium=0.95, zscore_threshold=3, min_group_size=20, low_volume_share_of_group_median=0.02, min_vc_for_outlier=0.25

Files: link_flows.csv rows 3609

SQL: WITH ph(period, hours) AS (VALUES ('AM', 3.0), ('PM', 3.0)), f AS (SELECT f.link_id, f.period, f.volume, COALESCE(f.vc_ratio, f.volume / NULLIF(l.capacity_vph * ph.hours, 0)) AS vc_ratio, l.link_class, l.area_type, l.capacity_vph, l.lanes, f.source_file, f.source_row FROM link_flows f JOIN links l USING (link_id) JOIN ph ON ph.period = f.period WHERE f.user_class = 'ALL' AND f.volume IS NOT NULL AND l.link_class NOT IN ('CONN')), g AS (SELECT link_class, area_type, period, COUNT(*) AS n, MEDIAN(LN(volume + 1)) AS med_log, MEDIAN(volume) AS med_vol FROM f GROUP BY 1, 2, 3), d AS (SELECT f.*, g.n, g.med_log, g.med_vol, LN(f.volume + 1) - g.med_log AS dev FROM f JOIN g USING (link_class, area_type, period)), m AS (SELECT link_class, area_type, period, MEDIAN(ABS(dev)) AS mad FROM d GROUP BY 1, 2, 3), scored AS (SELECT d.*, m.mad, CASE WHEN d.n >= 20 AND m.mad > 0 THEN d.dev / (1.4826 * m.mad) END AS zscore, (d.n >= 20 AND d.capacity_vph > 0 AND d.med_vol > 0 AND d.volume < 0.02 * d.med_vol) AS too_low FROM d JOIN m USING (link_class, area_type, period)) SELECT * FROM scored WHERE vc_ratio >= 0.95 OR (ABS(zscore) >= 3.0 AND COALESCE(vc_ratio, 0) >= 0.25) OR too_low ORDER BY vc_ratio DESC NULLS LAST, ABS(zscore) DESC NULLS LAST

[src: findings/ccd4663a9044; link_flows.csv (link_flows) row 3609 col volume]

Likely cause: Attribute mis-coding (class, speed or capacity) making the link unusually attractive or unattractive.

[src: findings/ccd4663a9044; link_flows.csv (link_flows) row 3609 col volume]

Suggested action: Review link attributes against neighbouring links of the same class.

[src: findings/ccd4663a9044; link_flows.csv (link_flows) row 3609 col volume]

Evidence

Metric	Value	Threshold
issue	statistical_outlier	

Metric	Value	Threshold
volume	880	
vc_ratio	0.59	
zscore	3.012	
group_median_volume	70	
group_size	48	
link_class	LOC	
area_type	RUR	
capacity_vph	500	
lanes	1	
too_low	no	
vc_critical		1.30
vc_high		1.10
vc_medium		0.95
zscore_threshold		3
min_group_size		20
low_volume_share_of_group_median		2.0%
min_vc_for_outlier		0.25

[src: findings/ccd4663a9044; link_flows.csv (link_flows) row 3609 col volume]

Sources: link_flows.csv (link_flows) row 3609 col volume

[src: findings/ccd4663a9044; link_flows.csv (link_flows) row 3609 col volume]

Demand matrix sanity: Zone 31, sector S3

Medium: 40.0% of trips starting in zone 31 stay inside it, above the 15% ceiling.

Modeller view: Method: intrazonal / row total per origin zone (all DEMAND matrices)

Values: issue=intrazonal_share, intrazonal_share=0.4004, intrazonal_trips=2568, total_trips=6413

Thresholds: intrazonal_share_max=0.15, min_zone_total_for_ratio=50

Files: od.csv

SQL: SELECT origin AS zone_id, SUM(trips) AS total, SUM(CASE WHEN origin = destination THEN trips ELSE 0 END) AS intrazonal, SUM(CASE WHEN origin = destination THEN trips ELSE 0 END) / NULLIF(SUM(trips), 0) AS share FROM od WHERE matrix_kind = 'DEMAND' GROUP BY 1 HAVING share > 0.15 AND SUM(trips) >= 50.0 ORDER BY share DESC, zone_id

[src: findings/d8aadcddee2b1; od.csv (od) col trips]

Likely cause: Intrazonal impedance far too low for this zone, or the zone is very large and should be split.

[src: findings/d8aadcddee2b1; od.csv (od) col trips]

Suggested action: Check the zone's intrazonal time/distance and its size.

[src: findings/d8aadcddee2b1; od.csv (od) col trips]

Evidence

Metric	Value	Threshold
issue	intrazonal_share	
intrazonal_share	40.0%	
intrazonal_trips	2,570	
total_trips	6,410	
intrazonal_share_max		15.0%
min_zone_total_for_ratio		50.00

[src: findings/d8aadcddee2b1; od.csv (od) col trips]

Sources: od.csv (od) col trips

[src: findings/d8aadcdde2b1; od.csv (od) col trips]

Centroid connector placement: Link 90003 (CONN, 1100-16)

Medium: Connector 90003 is 9.5 km long, above the 5.0 km limit.

Modeller view: Method: connector length_m above threshold

Values: issue=connector_too_long, length_m=9483, centroid=16

Thresholds: max_connector_length_m=5000

Files: links.csv rows 970

SQL: WITH conn AS (SELECT l.link_id, l.a_node, l.b_node, l.length_m, l.link_class, l.source_file, l.source_row, CASE WHEN COALESCE(na.is_centroid, FALSE) THEN l.b_node ELSE l.a_node END AS street_node, CASE WHEN COALESCE(na.is_centroid, FALSE) THEN l.a_node WHEN COALESCE(nb.is_centroid, FALSE) THEN l.b_node END AS centroid FROM links l LEFT JOIN nodes na ON na.node_id = l.a_node LEFT JOIN nodes nb ON nb.node_id = l.b_node WHERE l.link_class IN ('CONN') OR COALESCE(na.is_centroid, FALSE) OR COALESCE(nb.is_centroid, FALSE)), touch AS (SELECT c.link_id, MIN(h.link_id) AS high_link, ANY_VALUE(h.link_class) AS high_class FROM conn c JOIN links h ON (h.a_node = c.street_node OR h.b_node = c.street_node) AND h.link_class IN ('FWY', 'EXP') GROUP BY c.link_id) SELECT c.*, t.high_link, t.high_class FROM conn c LEFT JOIN touch t USING (link_id) WHERE t.high_link IS NOT NULL OR c.length_m > 5000.0 ORDER BY c.link_id

[src: findings/e002b66ba149; links.csv (links) row 970 col link_id]

Likely cause: Zone centroid far from the loaded network, or the connector length was left at a default.

[src: findings/e002b66ba149; links.csv (links) row 970 col link_id]

Suggested action: Split the zone, add local streets, or connect to a closer node.

[src: findings/e002b66ba149; links.csv (links) row 970 col link_id]

Evidence

Metric	Value	Threshold
issue	connector_too_long	
length_m	9,483	
centroid	16	
max_connector_length_m		5,000

[src: findings/e002b66ba149; links.csv (links) row 970 col link_id]

Sources: links.csv (links) row 970 col link_id

[src: findings/e002b66ba149; links.csv (links) row 970 col link_id]

Link volume outliers: Link 393 (LOC, 1016-1030), AM

Medium: Link 393 (LOC, RUR) carries 1,035 vehicles in the AM peak, far above similar links (typical 65).

Modeller view: Method: V/C from link_flows; robust $z = (\ln(v+1) - \text{median}) / (1.4826 \text{ MAD})$ within link_class x area_type x period

Values: issue=statistical_outlier, volume=1035, vc_ratio=0.69, zscore=3.197,

group_median_volume=65.14, group_size=48, link_class=LOC, area_type=RUR, capacity_vph=500, lanes=1, too_low=False

Thresholds: vc_critical=1.3, vc_high=1.1, vc_medium=0.95, zscore_threshold=3, min_group_size=20, low_volume_share_of_group_median=0.02, min_vc_for_outlier=0.25

Files: link_flows.csv rows 3297

SQL: WITH ph(period, hours) AS (VALUES ('AM', 3.0), ('PM', 3.0)), f AS (SELECT f.link_id, f.period, f.volume, COALESCE(f.vc_ratio, f.volume / NULLIF(l.capacity_vph * ph.hours, 0)) AS vc_ratio, l.link_class, l.area_type, l.capacity_vph, l.lanes, f.source_file, f.source_row FROM link_flows f JOIN links l USING (link_id) JOIN ph ON ph.period = f.period WHERE f.user_class = 'ALL' AND f.volume IS NOT NULL AND l.link_class NOT IN ('CONN')), g AS (SELECT link_class, area_type, period, COUNT(*) AS n,


```

MEDIAN(LN(volume + 1)) AS med_log, MEDIAN(volume) AS med_vol FROM f GROUP BY 1, 2, 3 ), d AS
( SELECT f.*, g.n, g.med_log, g.med_vol, LN(f.volume + 1) - g.med_log AS dev FROM f JOIN g USING
(link_class, area_type, period) ), m AS ( SELECT link_class, area_type, period, MEDIAN(ABS(dev)) AS
mad FROM d GROUP BY 1, 2, 3 ), scored AS ( SELECT d.*, m.mad, CASE WHEN d.n >= 20 AND m.mad
> 0 THEN d.dev / (1.4826 * m.mad) END AS zscore, (d.n >= 20 AND d.capacity_vph > 0 AND d.med_vol >
0 AND d.volume < 0.02 * d.med_vol) AS too_low FROM d JOIN m USING (link_class, area_type, period) )
SELECT * FROM scored WHERE vc_ratio >= 0.95 OR (ABS(zscore) >= 3.0 AND COALESCE(vc_ratio, 0)
>= 0.25) OR too_low ORDER BY vc_ratio DESC NULLS LAST, ABS(zscore) DESC NULLS LAST

```

[src: findings/ec37afcb6720; link_flows.csv (link_flows) row 3297 col volume]

Likely cause: Attribute mis-coding (class, speed or capacity) making the link unusually attractive or unattractive.

[src: findings/ec37afcb6720; link_flows.csv (link_flows) row 3297 col volume]

Suggested action: Review link attributes against neighbouring links of the same class.

[src: findings/ec37afcb6720; link_flows.csv (link_flows) row 3297 col volume]

Evidence

Metric	Value	Threshold
issue	statistical_outlier	
volume	1,030	
vc_ratio	0.69	
zscore	3.197	
group_median_volume	70	
group_size	48	
link_class	LOC	
area_type	RUR	
capacity_vph	500	
lanes	1	
too_low	no	
vc_critical		1.30
vc_high		1.10
vc_medium		0.95
zscore_threshold		3
min_group_size		20
low_volume_share_of_group_median		2.0%
min_vc_for_outlier		0.25

[src: findings/ec37afcb6720; link_flows.csv (link_flows) row 3297 col volume]

Sources: link_flows.csv (link_flows) row 3297 col volume

[src: findings/ec37afcb6720; link_flows.csv (link_flows) row 3297 col volume]

Info findings (0)

No Info findings for this run.

Convergence and noise

Check status: ok; findings: 1; message: noise band written for 4870 link/sector-period rows.

[src: check_results.json]

Convergence findings

Severity	Location	Finding
Critical	HWY_ASSIGN AM	The highway assignment for the AM peak stopped at a relative gap of 2.00% after 20 iterations, above the 0.10% target.

[src: findings/db2b60bb79ca]

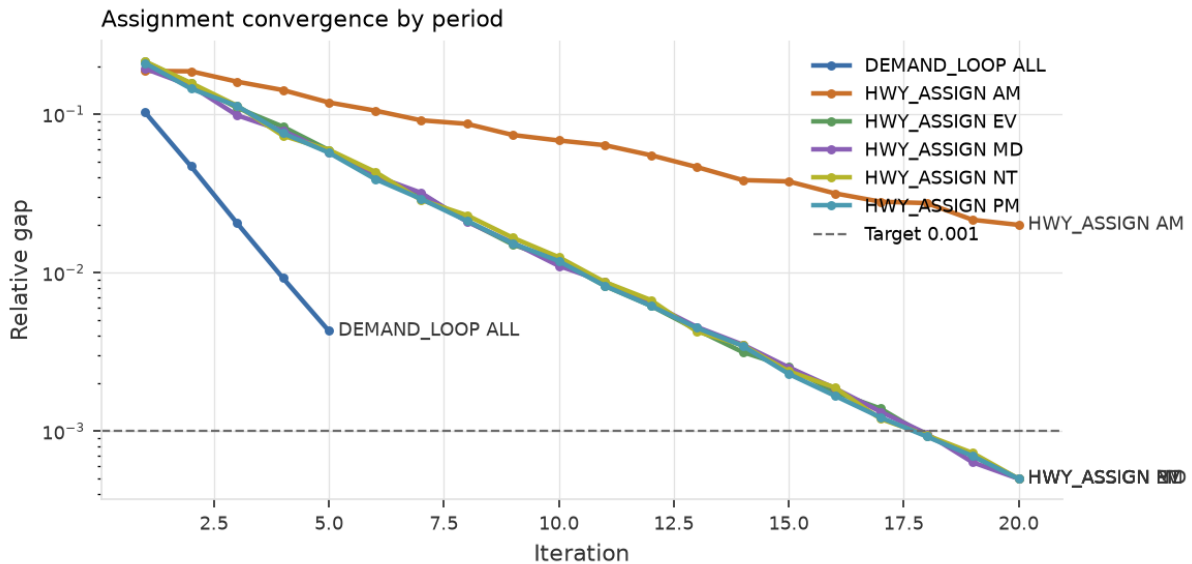
Final convergence values (relative gap target 0.001)

Stage	Period	Metric	Iterations	Final value	Target met
DEMAND_LOOP	ALL	REL_GAP	5	0.004283	no
HWY_ASSIGN	AM	GEH_PREV_ITER	20	2.01	
HWY_ASSIGN	AM	MAX_FLOW_CHANGE	20	555	
HWY_ASSIGN	AM	REL_GAP	20	0.02	no
HWY_ASSIGN	EV	GEH_PREV_ITER	20	0.2197	
HWY_ASSIGN	EV	MAX_FLOW_CHANGE	20	16.7	
HWY_ASSIGN	EV	REL_GAP	20	0.0005	yes
HWY_ASSIGN	MD	GEH_PREV_ITER	20	0.2197	
HWY_ASSIGN	MD	MAX_FLOW_CHANGE	20	35.2	
HWY_ASSIGN	MD	REL_GAP	20	0.0005	yes
HWY_ASSIGN	NT	GEH_PREV_ITER	20	0.2197	
HWY_ASSIGN	NT	MAX_FLOW_CHANGE	20	9.968	
HWY_ASSIGN	NT	REL_GAP	20	0.0005	yes
HWY_ASSIGN	PM	GEH_PREV_ITER	20	0.2197	
HWY_ASSIGN	PM	MAX_FLOW_CHANGE	20	30.0	
HWY_ASSIGN	PM	REL_GAP	20	0.0005	yes

[src: convergence table; config/checks.yaml]

The noise band table has 4870 rows with columns level, location_id, period, metric, band_low, band_high, method.

[src: noise_band.parquet]



Relative gap by iteration and period (log scale)

[src: convergence table (metric REL_GAP); config/checks.yaml rel_gap_target]

Key performance indicators

KPIs with their published definitions

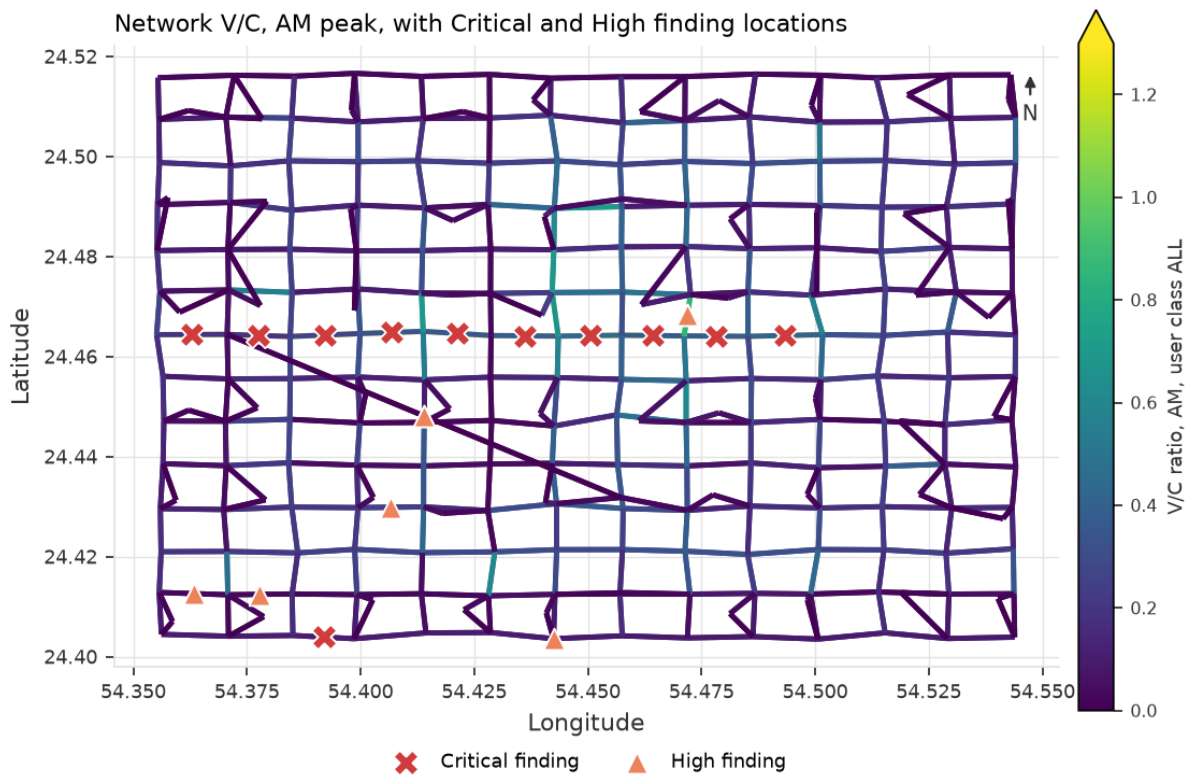
KPI	Value	Period	Definition
Vehicle-km travelled, AM	1,751,400 veh-km	AM	Sum of link volume x length for user_class ALL in the AM period. SQL: SELECT SUM(f.volume * l.length_m / 1000.0) AS value FROM link_flows f JOIN links l USING (link_id) WHERE f.period = 'AM' AND f.user_class = 'ALL'
Share of links with V/C above 1.0, AM	1.3%	AM	Links with vc_ratio > 1.0 divided by all links carrying flow, AM, user_class ALL. SQL: SELECT AVG(CASE WHEN vc_ratio > 1.0 THEN 1.0 ELSE 0.0 END) AS value FROM link_flows WHERE period = 'AM' AND user_class = 'ALL' AND volume > 0
Volume-weighted congested speed, AM	73 km/h	AM	Sum(volume x congested speed) / Sum(volume) for AM, user_class ALL. SQL: SELECT SUM(volume * cong_speed_kph) / NULLIF(SUM(volume), 0) AS value FROM link_flows WHERE period = 'AM' AND user_class = 'ALL'
Total demand trips, AM	93,250 trips	AM	Sum of all DEMAND matrix cells in the AM period, all purposes and modes. SQL: SELECT SUM(trips) AS value FROM od WHERE matrix_kind = 'DEMAND' AND period = 'AM'
PT mode share, AM	14.3%	AM	PT demand trips divided by all demand trips, AM. SQL: SELECT SUM(CASE WHEN mode = 'PT' THEN trips ELSE 0 END) / NULLIF(SUM(trips), 0) AS value FROM od WHERE matrix_kind = 'DEMAND' AND period = 'AM'
Mean trip length, all periods	6.7 km	all	Trip-weighted mean of the DIST skim (metres, per data contract 8.3) over DEMAND matrix cells, converted to km. SQL: SELECT SUM(o.trips * s.value) / NULLIF(SUM(o.trips), 0) / 1000.0 AS value FROM od o JOIN skims s ON s.skim_kind = 'DIST' AND s.mode = o.mode AND s.period = o.period AND s.origin = o.origin AND s.destination = o.destination WHERE o.matrix_kind = 'DEMAND'
Road network length	1096.4 km	all	Sum of link length over all highway links (each direction counted). SQL: SELECT SUM(length_m) / 1000.0 AS value FROM links
Lane-km, all roads	1,790 lane-km	all	Sum of link length x lanes over all highway links. SQL: SELECT SUM(length_m * lanes) / 1000.0 AS value FROM links
Lane-km, freeways	567 lane-km	all	Sum of link length x lanes over links of class FWY. SQL: SELECT SUM(length_m * lanes) / 1000.0 AS value FROM links WHERE link_class = 'FWY'

KPI	Value	Period	Definition
Total demand trips, all periods	393,810 trips	all	Sum of all DEMAND matrix cells, every purpose, mode and period. SQL: SELECT SUM(trips) AS value FROM od WHERE matrix_kind = 'DEMAND'
Population	283,630 people	all	Sum of the POP land-use variable over all zones. SQL: SELECT SUM(value) AS value FROM land_use WHERE variable = 'POP'

[src: kpis.json; config/kpis.yaml]

DRAFT

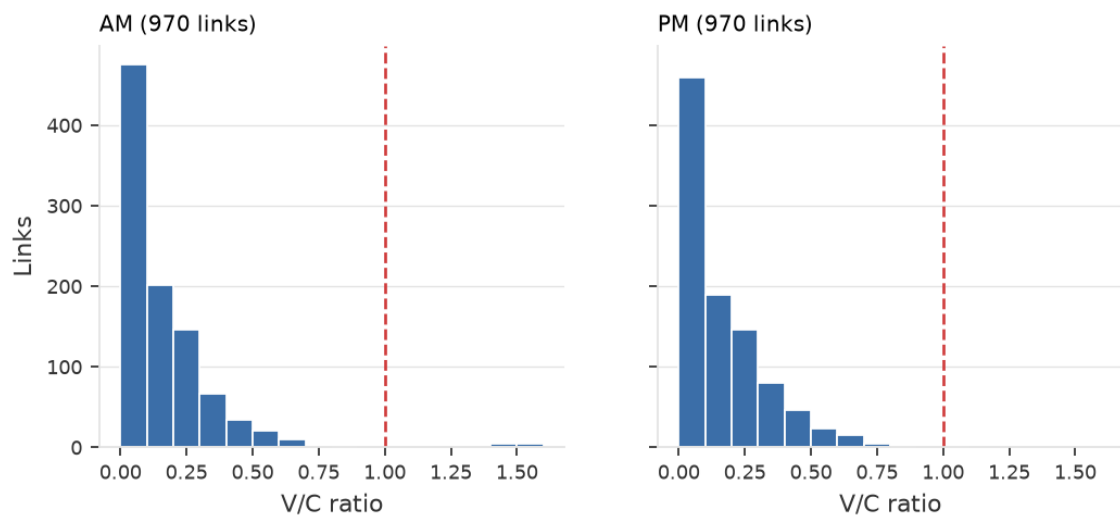
Maps and charts



Network map: links coloured by AM V/C (user class ALL), Critical and High finding locations marked

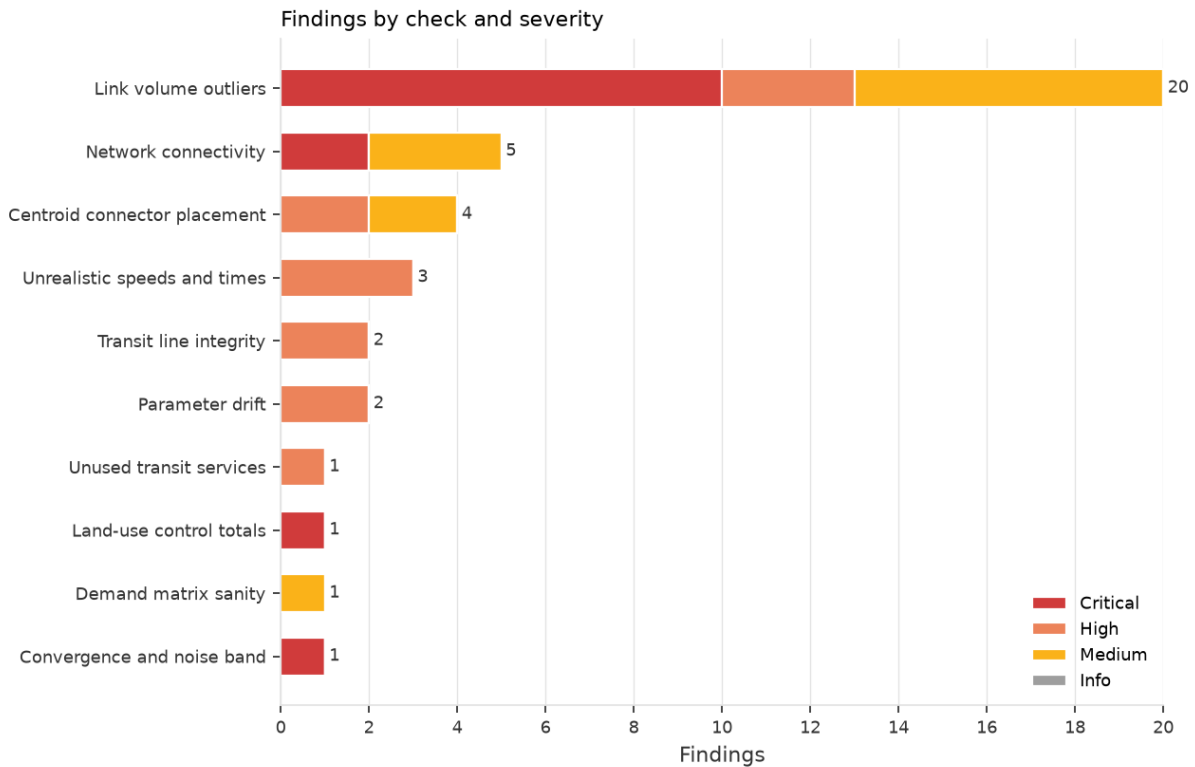
[src: links table (geometry_wkt); link_flows table period AM user_class ALL; findings.json]

Link V/C distribution, user class ALL



Distribution of link V/C ratios, AM and PM (user class ALL)

[src: link_flows table; user_class ALL; periods AM, PM]



Findings by check and severity

[src: findings.json; 40 findings]

Check coverage

13 checks were run: 11 completed, 2 skipped because a required table was missing, 0 ended in error.

[src: check_results.json]

Check coverage

Check	Status	Findings	Message	Duration (s)
Null values and identifier integrity	ok	0		0.15
Network connectivity	ok	5		0.03
Centroid connector placement	ok	4		0.01
Transit line integrity	ok	2		0.02
Land-use control totals	ok	1		0.01
Link volume outliers	ok	20		0.04
Convergence and noise band	ok	1	noise band written for 4870 link/sector-period rows	0.07
Demand matrix sanity	ok	1		0.06
Trip length distribution	skipped	0	needs a base run and none is linked to this run	0.00
Unrealistic speeds and times	ok	3		0.01
Parameter drift	ok	2		0.01
Unexplained demand shift	skipped	0	needs a base run and none is linked to this run	0.00
Unused transit services	ok	1		0.03

[src: check_results.json]

Appendix

Method notes

Each check maps a computed score to a severity band. Bands are checked top-down and the first match wins. The thresholds below are read from config/checks.yaml (defaults: enabled=True, max_findings_per_check=200, group_issue_above=25, group_keep_worst=10).

[src: config/checks.yaml]

Check thresholds and severity rules

Check	Enabled	Parameters	Severity rules
null_and_id_integrity	yes	required_columns = {'links': ['link_id', 'a_node', 'b_node', 'length_m', 'link_class', 'capacity_vph', 'ffs_kph'], 'nodes': ['node_id', 'x', 'y'], 'zones': ['zone_id'], 'link_flows': ['link_id', 'period', 'user_class', 'volume']}	Critical when missing_node_refs > 0; Critical when missing_zone_refs > 0; High when missing_link_refs > 0; High when null_share > 0.01; Medium when null_share > 0
network_connectivity	yes	min_capacity_vph = 1; min_ffs_kph = 5; touch_tolerance_m = 1; near_gap_m = 50	Critical when issue == 'zero_capacity' or issue == 'zero_speed'; High when issue == 'node_id_mismatch'; Medium when issue == 'orphan_node'; High when issue == 'oneway_dead_end'; High when issue == 'disconnected_component'
centroid_connectors	yes	high_class_links = FWY, EXP; connector_classes = CONN; max_connector_length_m = 5,000	High when issue == 'connector_on_high_class'; Medium when issue == 'connector_too_long'
transit_line_integrity	yes	headway_min_range = 2, 120; modes_headway_max = {'METRO': 20, 'LRT': 30, 'BRT': 30, 'BUS': 120, 'RAIL': 120, 'TRAM': 30, 'FERRY': 120}	Critical when issue == 'broken_sequence'; High when issue == 'headway_out_of_range'; Critical when issue == 'node_not_in_network'
land_use_control_totals	yes	tolerance_share = 0.02	Critical when abs_diff_share > 0.10; High when abs_diff_share > 0.05; Medium when abs_diff_share > 0.02
link_volume_outliers	yes	vc_critical = 1.3; vc_high = 1.1; vc_medium = 0.95; zscore_threshold = 3; min_vc_for_outlier = 0.25; min_group_size = 20; low_volume_share_of_group_median = 0.02; user_class = ALL; periods = AM, PM; exclude_link_classes = CONN	Critical when vc_ratio >= 1.30; High when vc_ratio >= 1.10; High when abs_zscore >= 4.0; Medium when vc_ratio >= 0.95 or abs_zscore >= 3.0; Medium when too_low
convergence_and_noise	yes	rel_gap_target = 0.001; rel_gap_high = 0.01; geh_stable = 1; geh_stable_share_target = 0.95; noise_band_iterations = 3; stages = HWY_ASSIGN, PT_ASSIGN	Critical when rel_gap > 0.01; High when rel_gap > 0.001; Medium when stable_share < 0.95
matrix_sanity	yes	intrazonal_share_max = 0.15; row_col_ratio_max = 5; min_zone_total_for_ratio = 50; pa_format_share = 0.25; fractional_cell_max_trips = 0.01; fractional_share_min = 0.01	Critical when issue == 'negative_cells'; High when issue == 'row_col_imbalance'; Medium when issue == 'intrazonal_share'; Info when issue == 'fractional_artefact'
trip_length_distribution	yes	max_mean_diff_share = 0.15; max_mean_diff_share_high = 0.3; distance_bins_km = 0, 2, 5, 10, 20, 40, 80, 200; skim_dist_units = km; reference_mean_km = {}	High when mean_diff_share > 0.30; Medium when mean_diff_share > 0.15
unrealistic_speeds_times	yes	min_cong_speed_kph = 5; max_speed_over_ffs_ratio = 1.05; max_delay_s = 900; periods = AM, PM; user_class = ALL	High when issue == 'speed_above_ffs'; High when issue == 'speed_below_floor'; Medium when issue == 'delay_outlier'
parameter_drift	yes	numeric_tolerance = 1e-09	High when drifted
unexplained_demand_and_shift	yes	sector_shift_share = 0.1; catchment_km = 5; min_sector_pair_trips = 100; explain_land_use_share = 0.01	High when shift_share > 0.25; Medium when shift_share > 0.10
unused_transit_services	yes	min_load_factor = 0.05; min_catchment_demand = 500; catchment_km = 1; pt_modes = PT	High when load_factor < 0.02; Medium when load_factor < 0.05

[src: config/checks.yaml]

Glossary

Glossary

Term	Meaning
V/C	Volume-to-capacity ratio: link volume per hour divided by link capacity per hour. Values above 1.0 mean demand exceeds capacity.
GEH	A statistic comparing two flows, M and C: $\sqrt{2(M - C)^2 / (M + C)}$. Values below 5 are usually treated as a good match.
Relative gap	Assignment convergence measure: the excess cost of the current route choices over the cheapest possible routes, divided by total cost. Smaller is better.
Noise band	The size of link flow change between the last assignment iterations. Differences inside the band are not treated as significant.
Centroid connector	An artificial link that loads a zone's trips onto the network at a chosen node.
Screenline	An imaginary line across the network on which modelled and observed crossings are compared.

[src: STEAM-AI glossary]